

SERVICE DATA
FILE NO. 300-763

TOSHIBA

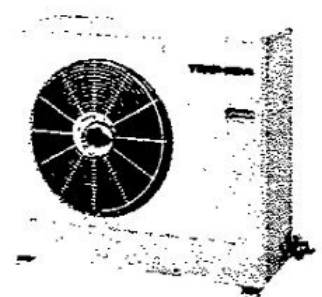
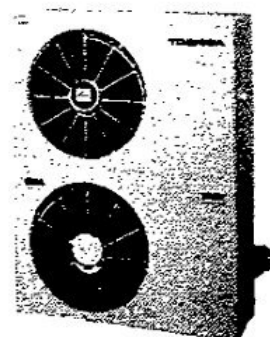
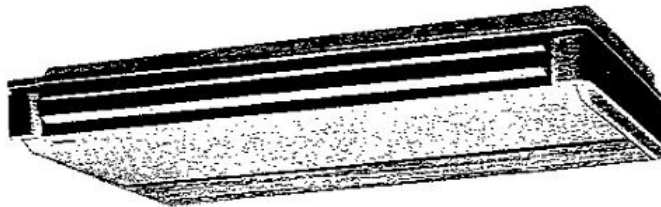
AIR-CONDITIONER SPLIT (CEILING TYPE) HEAT PUMP

RAV-453CHE/S453HE

RAV-713CHE8/S713HE8

RAV-1003CHE8/S1003HE8

RAV-1253CHE8/S1253HE8



Specifications are subject to change without notice.

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1. SPECIFICATIONS

ITEM	MODEL	RAV-453CHE	RAV-713CHE8
Cooling capacity	*1 kcal/h	4,000	6,300
	BTU/h	16,000	25,200
	kW	4.7	7.3
Heating capacity	*2 kcal/h	4,300	6,800
	BTU/h	17,200	27,200
	kW	5.0	7.9
Power source	Phase	1	3
	V	220/240	380/415
	Hz	50	50
Power consumption	kW	COOLING 2.0	COOLING 2.6
Power factor	%	90	85
Running current	A	9.7	4.4
Starting current	A	60	28
Operating noise (SPL)	Indoor unit dB(A)	46/42/39	48/44/41
	Outdoor unit dB(A)	50	53
	Name of refrigerant	R-22	R-22
Refrigerant	Charge volume kg	1.35	2.3
	Add. volume (5-25m) g/m	20	35
Refrigerant control		Capillary tube & Expansion valve	Capillary tube & Expansion valve
Interconnection pipe	Gas side size mm (in.)	φ 12.7	φ 15.9
	Coupler style	Flare	Flare
	Liquid side size mm (in.)	φ 6.4	φ 9.5
	Coupler style	Flare	Flare
	Standard length m (ft)	5 (16.4')	5 (16.4')
	Maximum length *3 (of one way) m (ft)	25 (82')	25 (82')
	Maximum height		
	Indoor unit higher m (ft)	15 (49')	15 (49')
	Outdoor unit higher m (ft)	15 (49')	15 (49')
Condensate drain pipe diameter	mm	φ 20 (ID)	φ 20 (ID)
INDOOR UNIT		RAV-453CHE	RAV-713CHE8
Model			
Appearance colour		Silky white (Munsell 5Y8.9/1)	Silky white (Munsell 5Y8.9/1)
Dimensions	Height mm (ft-in.)	195 (7-3/4")	195 (7-3/4")
	Width mm (ft-in.)	1070 (3'6-1/8")	1270 (4'2")
	Depth mm (ft-in.)	600 (1'11-5/8")	600 (1'11-5/8")
Net weight	kg (lbs)	28 (62)	33 (73)
Heat exchanger type		Finned tube	Finned tube
Indoor fan type		Centrifugal fan	Centrifugal fan
Air volume	m³/h (CFM)	780 (459)	1,200 (706)
Fan motor output	W	46	75
Air filter		Washable (One touch detachable)	Washable (One touch detachable)
Auto louver		Provided	Provided
OUTDOOR UNIT		RAV-S453HE	RAV-S713HE8
Model			
Appearance colour		Bronze white (Munsell 6Y7.5/1)	Bronze white (Munsell 6Y7.5/1)
Dimensions	Height mm (ft-in.)	740 (2'5-1/8")	790 (2'7-1/8")
	Width mm (ft-in.)	880 (2'10-5/8")	880 (2'10-5/8")
	Depth mm (ft-in.)	310 (1'3/16")	310 (1'3/16")
Net weight	kg (lbs)	60 (132)	80 (176)
Heat exchanger type		Finned tube	Finned tube
Outdoor fan type		Propeller fan	Propeller fan
Fan motor output	W	39	63
Compressor	Model	PH230X3-4LS	YH277JA
	Output kW	1.5	2.0
Protective device		High pressure switch, Fuse Overcurrent relay, Crankcase heater inner over load relay	High pressure switch, Fuse Overcurrent relay, Crankcase heater inner over load relay

Specifications are subject to change without notice.

Note 1: Cooling capacity is based on the following temperature conditions.

Indoor air inlet temperature: 27°C DB (80°F DB)
19.5°C WB (67°F WB)
Outdoor air inlet temperature: 35°C DB (95°F DB)

Note 2: Heating capacity is based on the following temperature conditions.

Indoor air inlet temperature: 21°C DB (70°F DB)
Outdoor air inlet temperature: 7°C DB (45°F DB)
6°C WB (43°F WB)

Note 3: These mean actual length.

Note 4: Operating range of the units

Indoor air temperature

When cooling

Maximum 32°C DB, 22.5°C WB (90°F DB, 73°F WB)

Minimum 18°C DB, 15.5°C WB (65°F DB, 60°F WB)

When heating

Maximum 30°C DB (86°F DB)

Minimum 18°C DB (65°F DB)

Outdoor air temperature

When cooling

Maximum 43°C DB, 25.5°C WB (109°F DB, 78°F WB)

Minimum 10°C DB (50°F DB)

When heating

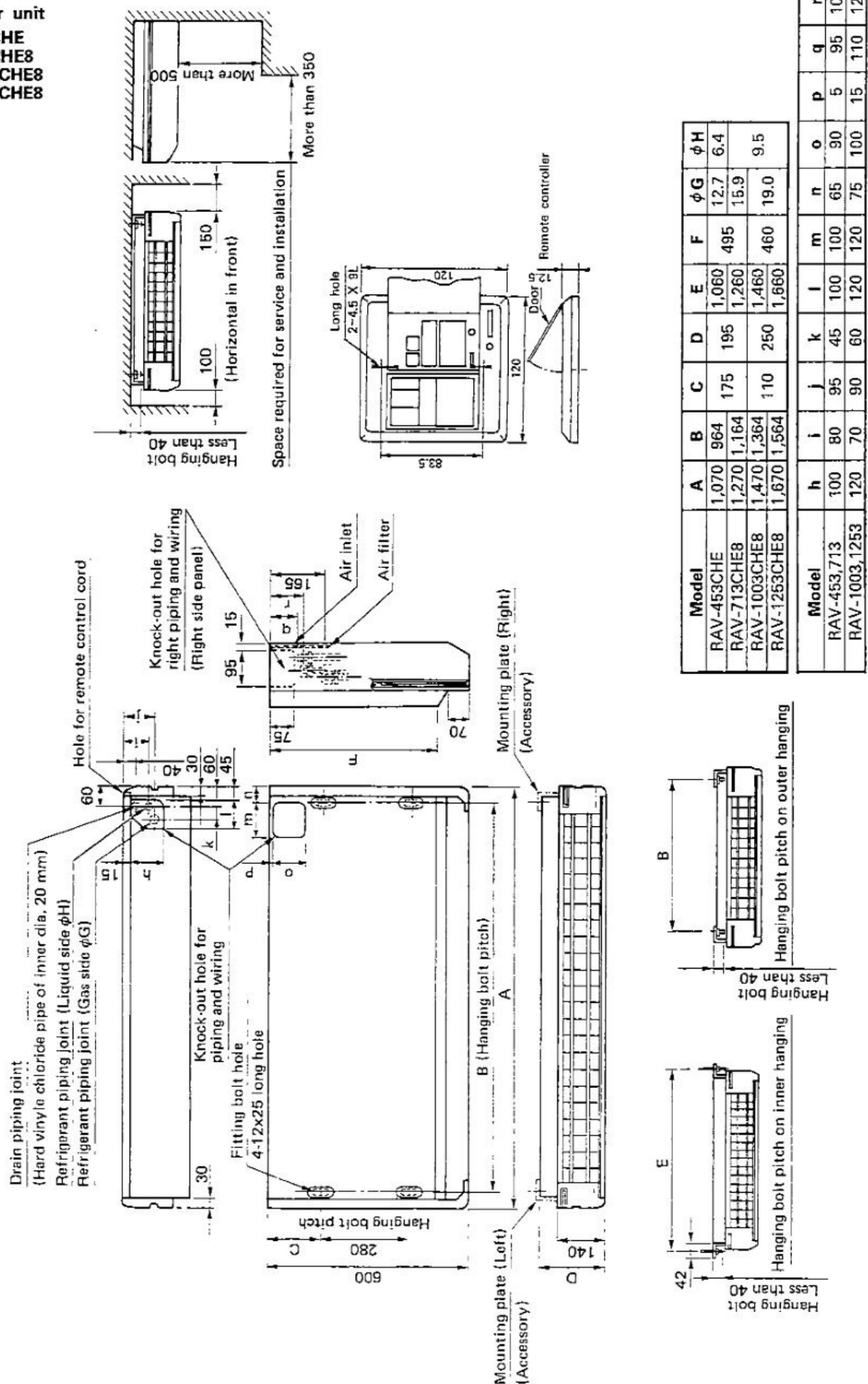
Maximum 21°C DB, 15.5°C WB (70°F DB, 60°F WB)

Minimum -10°C WB (14°F WB)

2. CONSTRUCTION VIEWS

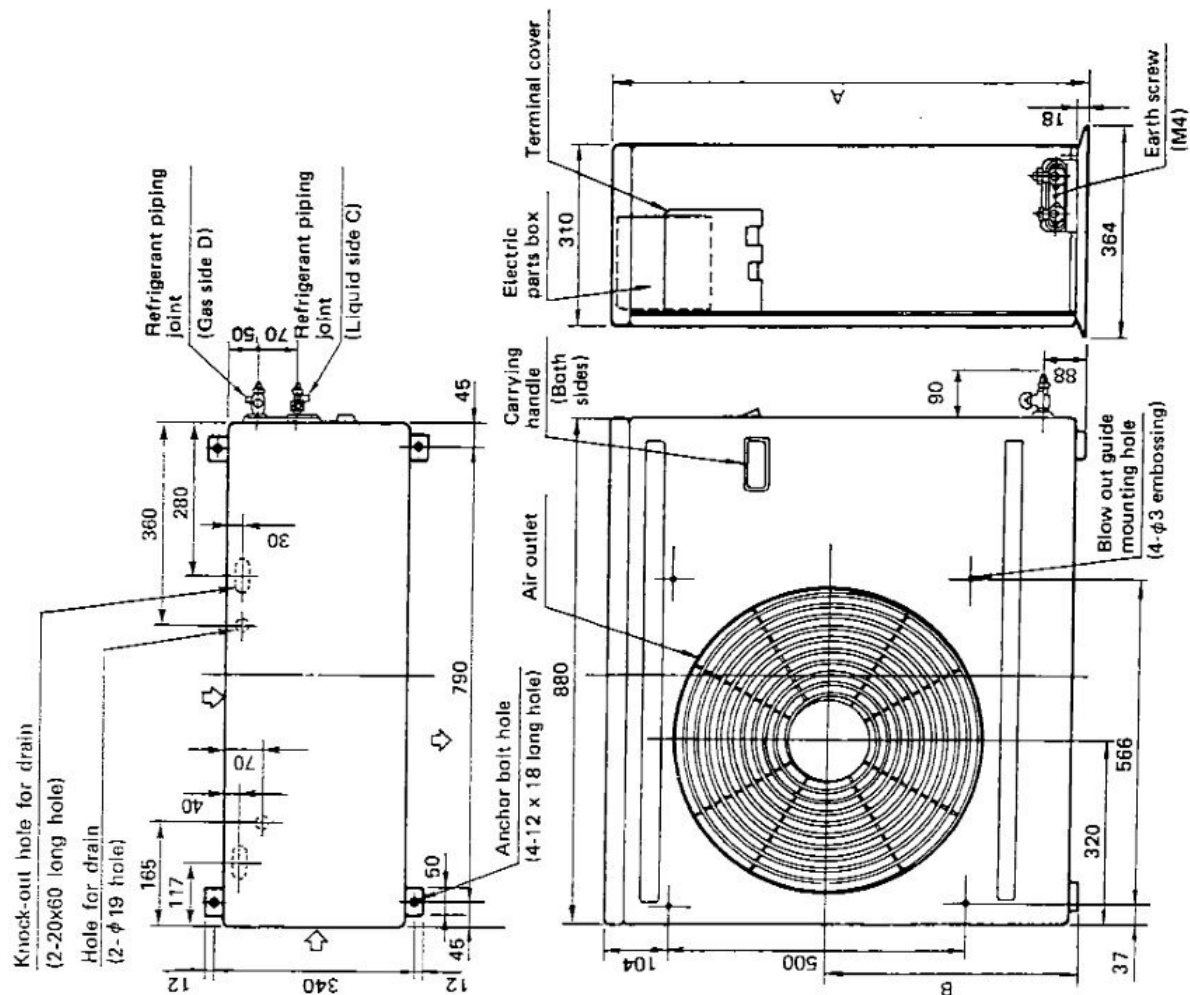
2.1 Indoor unit

RAV-453CHE
RAV-713CHE8
RAV-1003CHE8
RAV-1253CHE8

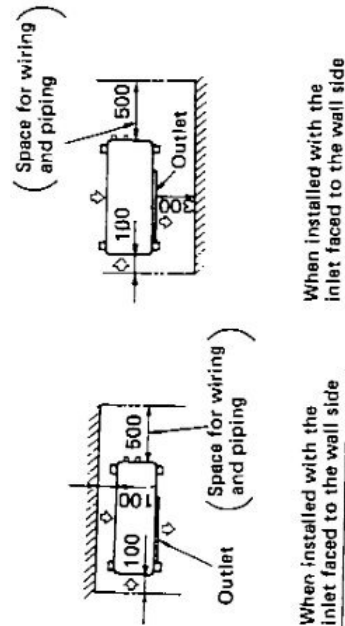


2.2 Outdoor unit

RAV-S453HE
RAV-S713HE8



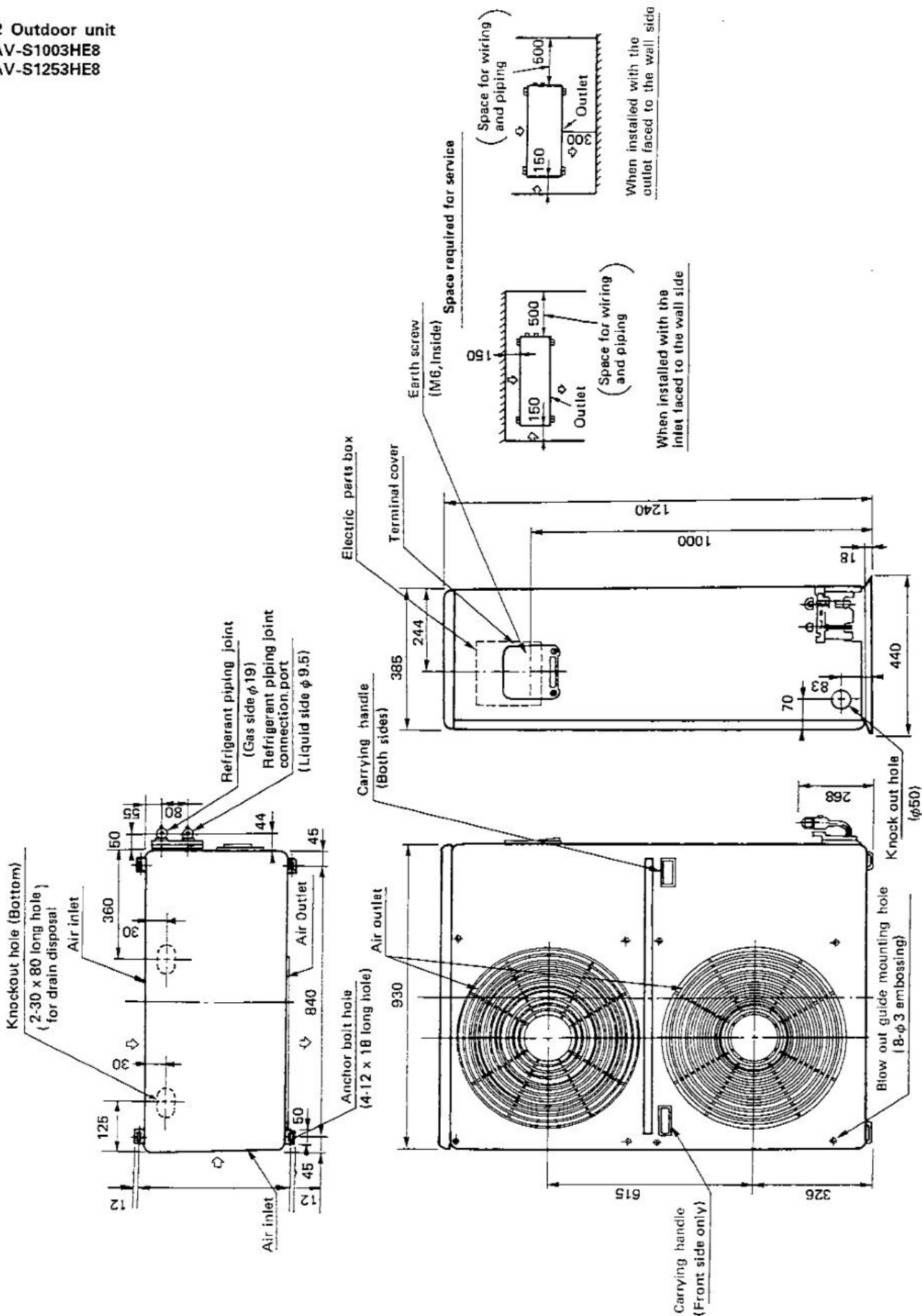
Space required for service



(Unit: mm)

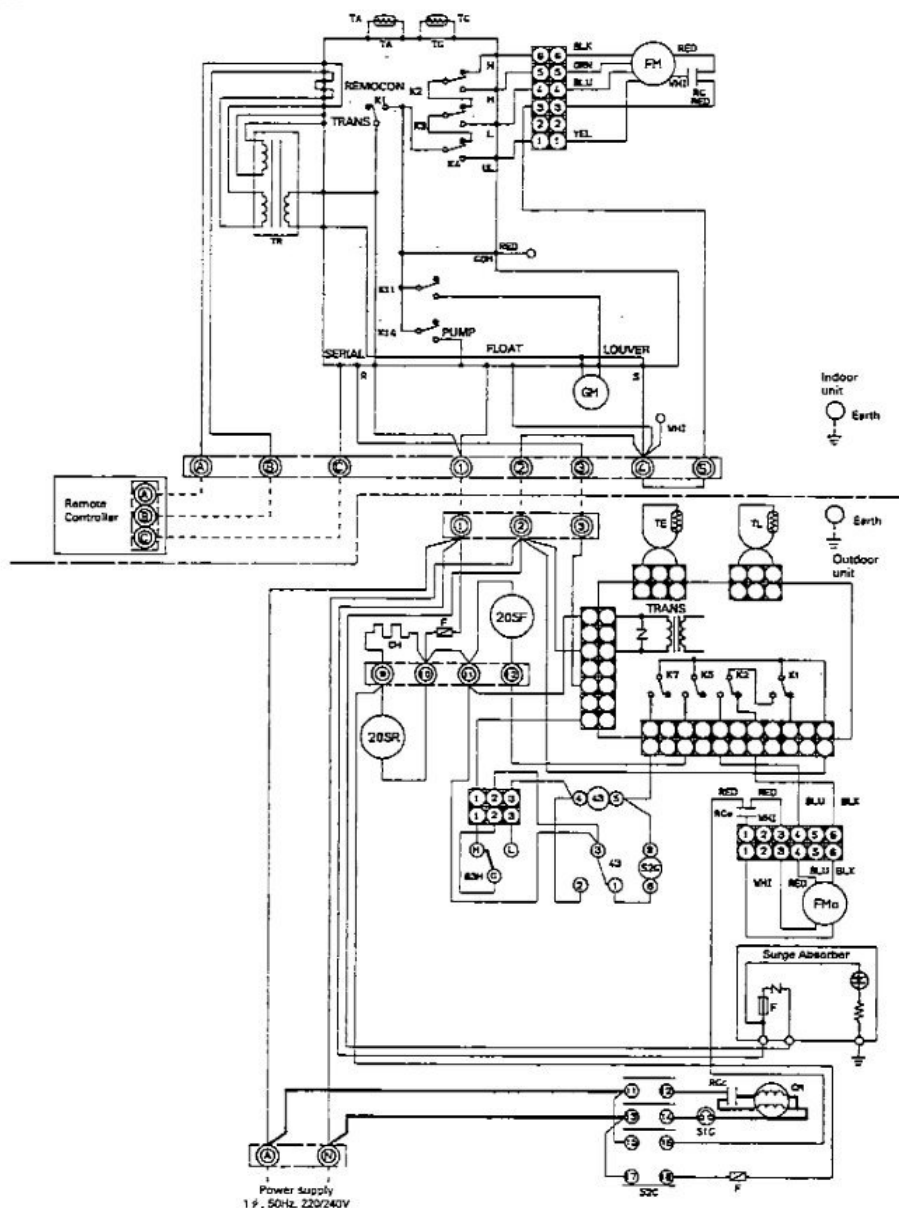
Model	A	B	C	D
RAV-S453HE	740	380	φ6.4	φ12.7
RAV-S713HE8	790	430	φ9.5	φ15.9

2.2 Outdoor unit
RAV-S1003HE8
RAV-S1253HE8



3. WIRING DIAGRAM

RAV-453CHE/S453HE

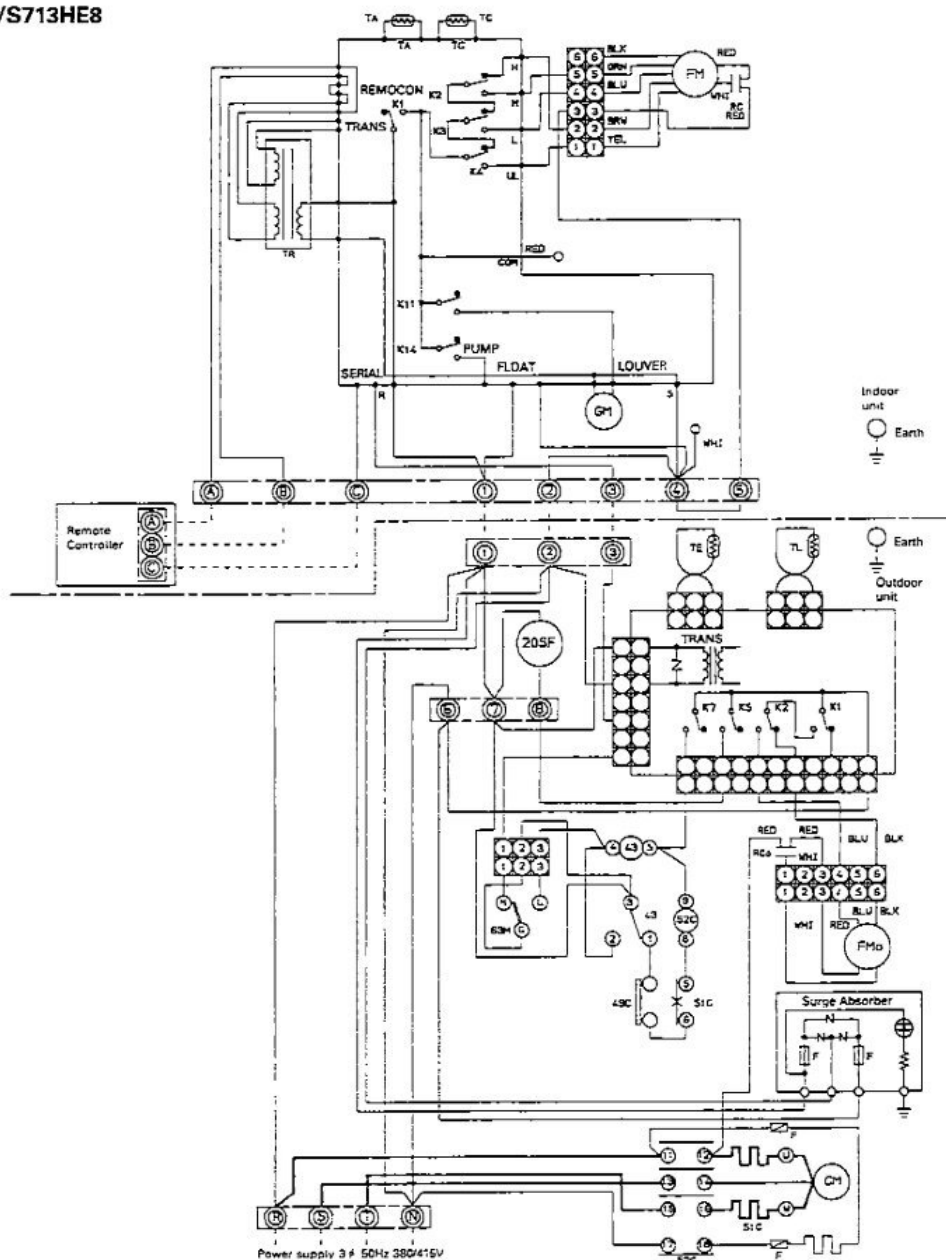


© shows terminal block and figures show terminal numbers.
Broken lines show wiring at site.

Don't operate the units with the magnetic contactor pushed.

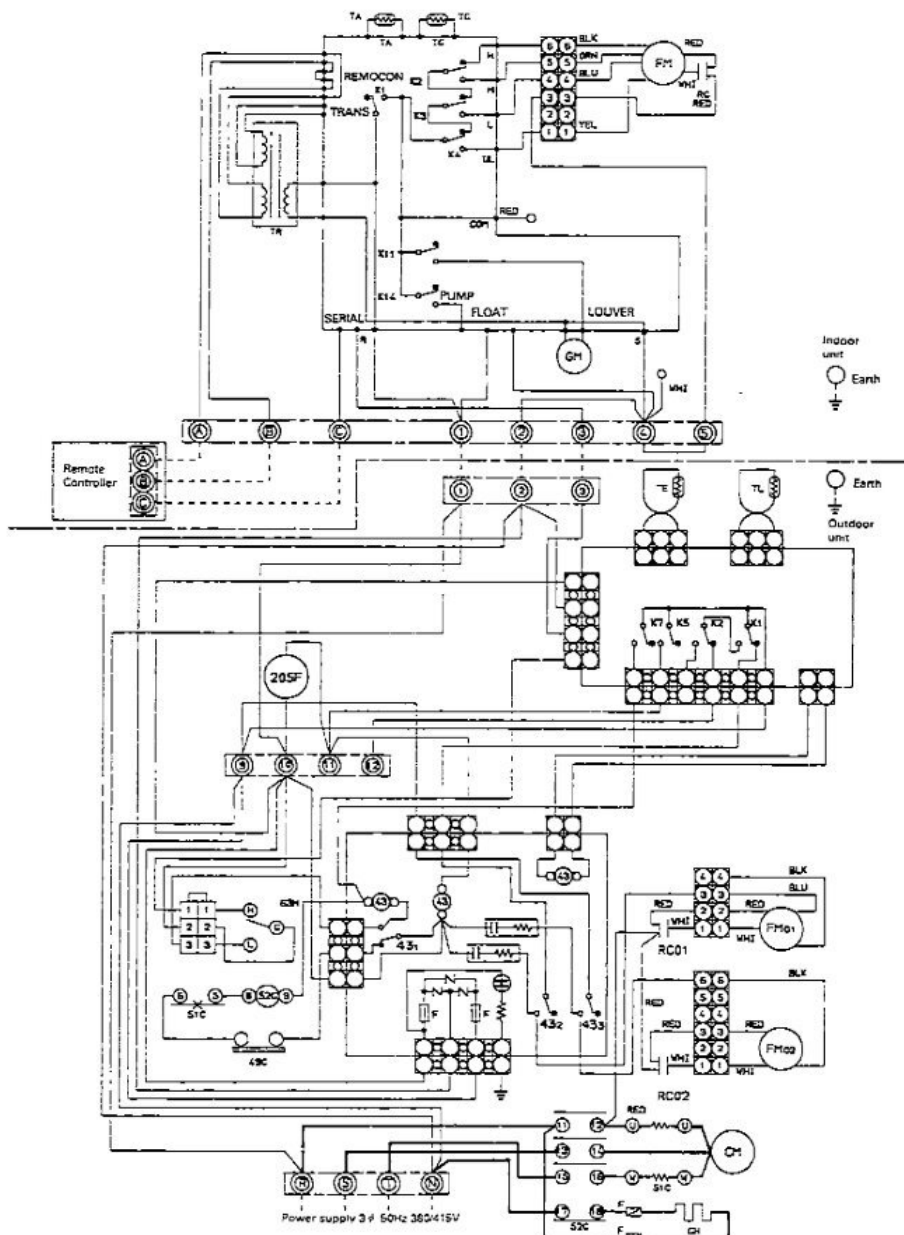
Symbol	Name	Symbol	Name
20SF	Solenoid Coil (4way valve)	CM	Compressor
K ₁ ~K ₁₄	Relay	52C	Magnetic Contactor
20SR	Solenoid Coil (2way valve)	FM _O	Fan Motor (Outdoor)
51C	Overload Relay	TL	Sensor
43	Relay (Outdoor)	TE	Sensor
63H	High Pressure Switch	F	Fuse
CH	Crank Case Heater	RC _O	Running Capacitor (Outdoor Fan Motor)
TA	Sensor	RC _C	Running Capacitor (Compressor)
TC	Sensor	RC	Running Capacitor (Indoor Fan Motor)
FM	Fan Motor (Indoor)		
GM	Induced Draft Motor		
TR	Transformer		

RAV-713CHE8/S713HE8



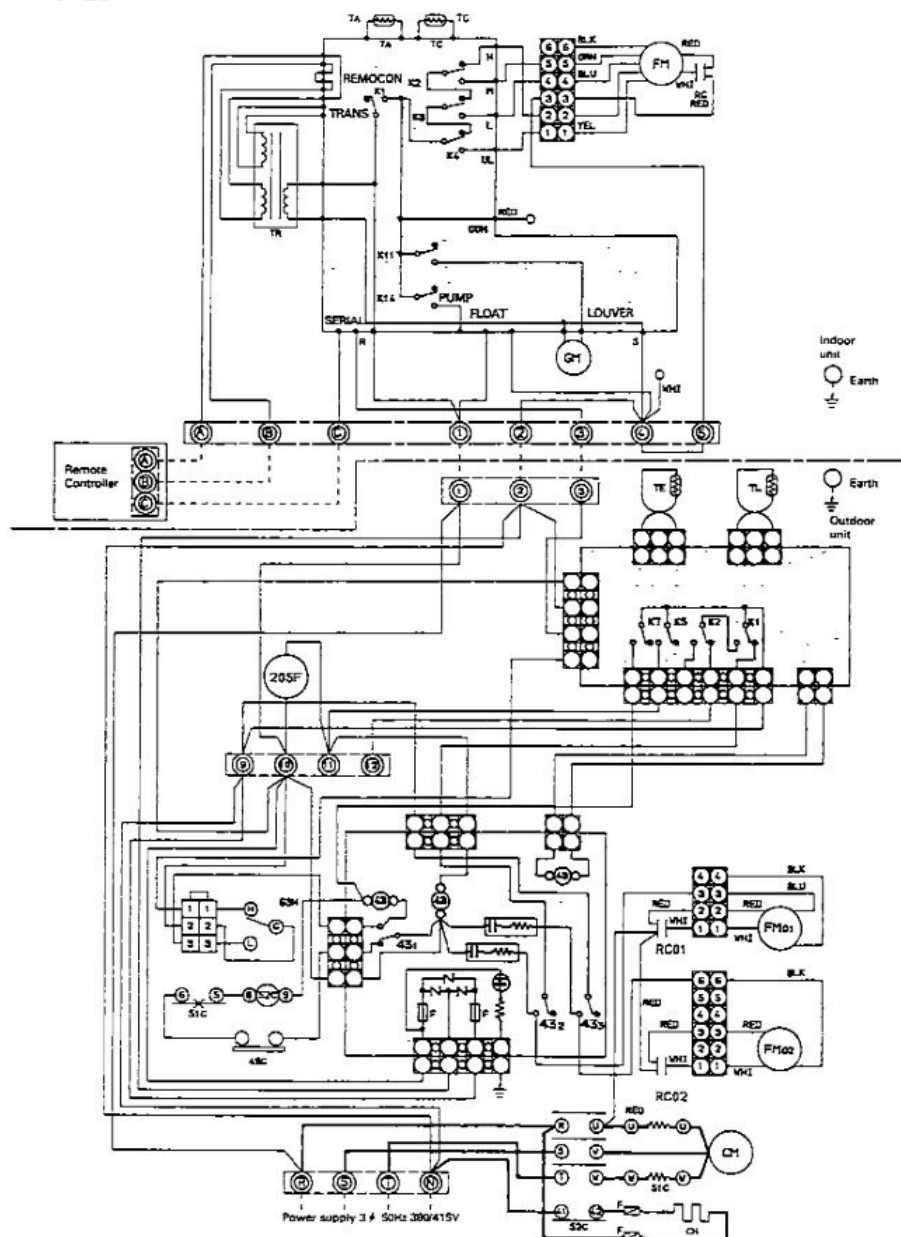
© shows terminal block and figures show terminal numbers.
Broken lines show wiring at site.

Don't operate the units with the magnetic contactor pushed.			
Symbol	Name	Symbol	Name
20SF	Solenoid Coil (4way valve)	CM	Compressor
K ₁ ~K ₁₄	Relay	52C	Magnetic Contactor
49C	Inner Overload Relay	FM _O	Fan Motor (Outdoor)
51C	Overload Relay	TL	Sensor
43	Relay (Outdoor)	TE	Sensor
63H	High Pressure Switch	F	Fuse
CH	Crank Case Heater	RC _O	Running Capacitor (Outdoor Fan Motor)
TA	Sensor	RC	Running Capacitor (Indoor Fan Motor)
TC	Sensor		
FM	Fan Motor (Indoor)		
GM	Induced Draft Motor	TR	Transformer

RAV-1003CHE8/S1003HE8


© shows terminal block and figures show terminal numbers.
Broken lines show wiring at site.

Don't operate the units with the magnetic contactor pushed.			
Symbol	Name	Symbol	Name
20SF	Solenoid Coil	CM	Compressor
K ₁ ~K ₁₄	Relay	52C	Magnetic Contactor
49C	Inner Overload Relay	FM ₀₁₋₀₂	Fan Motor (Outdoor)
51C	Overload Relay	TL	Sensor
43 ₁₋₃	Relay (Outdoor)	TE	Sensor
63H	High Pressure Switch	F	Fuse
CH	Crank Case Heater	RC ₀₁₋₀₂	Running Capacitor (Outdoor Fan Motor)
TA	Sensor	RC	Running Capacitor (Indoor Fan Motor)
TC	Sensor	TR	Transformer
GM	Induced Draft Motor		
FM	Fan Motor (Indoor)		

RAV-1253CHE8/S1253HE8


© shows terminal block and figures show terminal numbers.
Broken lines show wiring at site.

Don't operate the units with the magnetic contactor pushed.

Symbol	Name	Symbol	Name
20SF	Solenoid Coil	CM	Compressor
K ₁ ~K ₁₄	Relay	52C	Magnetic Contactor
49C	Inner Overload Relay	FM _{01,02}	Fan Motor (Outdoor)
51C	Overload Relay	TL	Sensor
43 _{1~3}	Relay (Outdoor)	TE	Sensor
63H	High Pressure Switch	F	Fuse
CH	Crank Case Heater	RC _{01,02}	Running Capacitor (Outdoor Fan Motor)
TA	Sensor	RC	Running Capacitor (Indoor Fan Motor)
TC	Sensor	TR	Transformer
GM	Induced Draft Motor		
FM	Fan Motor (Indoor)		

4. SPECIFICATIONS OF ELECTRICAL PARTS

4.1 RAV-453CHE/S453HE

NO.	PARTS NAME	TYPE	SPECIFICATIONS						
1	Indoor unit fan motor	AF-230-46-4D	Output (Rated) 46W, 4pole, 1phase, 230V, 50Hz						
2	Running capacitor for indoor fan motor	45FAFN305UJ	AC450V, 3 μ F						
3	Transformer	FT12	187~264V						
4	Induced draft motor	SGM-220-5A	AC220/240V						
5	Sensor for room temperature		Maximum input 450mW	°C	15	20	25	30	40
				k Ω	16.1	12.6	10.0	8.0	5.2
6	Indoor unit sensor for heat-exchanger temp.	DTN-C103J40	Maximum input 34mA	°C	-12	0	25	50	
				k Ω	62.29	32.82	10.0	3.59	
7	Compressor	PH230X3-4LS	Output (Rated) 1.5kW, 2pole, 1phase, 220/240V, 50Hz						
8	Running capacitor for compressor	MT-42MP456W	AC420V, 45 μ F						
9	Outdoor unit fan motor	AF-230-39P	Output (Rated) 39W, 6pole, 1phase, 230V, 50Hz						
10	Running capacitor for outdoor fan motor	35FAFN405UJ	AC350V, 4 μ F						
11	Magnetic contactor	E-20F-F	AC230V, 50Hz						
12	High pressure switch	HTB-T317	Tripping pressure 30kg/cm ² G Resetting pressure 23kg/cm ² G						
13	Solenoid coil for four way valve	L270021	AC240V, 50/60Hz						
14	Solenoid coil for two way valve		AC220/240V, 50Hz						
15	Relay	SA106H	AC230V, 50/60Hz						
16	Crankcase heater		AC240V, 28W						
17	Sensor for defrosting		Maximum input 15.5mA	°C	-12	0	25	50	
				k Ω	67.5	32.82	10.0	3.59	
18	Sensor for cooling operation in low ambient temperature	DTN-C103J40	Maximum input 34mA	°C	-12	0	25	50	
				k Ω	62.29	32.82	10.0	3.59	
19	Overload relay	OL-177GM15	AC240V, Tripping temp. 165°C, Resetting temp. 80°C						
20	Fuse		1A						

4.2 RAV-713CHE8/S713HE8

NO.	PARTS NAME	TYPE	SPECIFICATIONS						
1	Indoor unit fan motor	AF-200-75-4N	Output (Rated) 75W, 4pole, 1phase, 200V, 50/60Hz						
2	Running capacitor for indoor fan motor	EFG40M255UF7	AC400V, 2.5 μ F						
3	Transformer	FT12	187~264V						
4	Induced draft motor	SGM-220-5A	AC220/240V						
5	Sensor for room temperature		Maximum input 450mW	°C	15	20	25	30	40
				k Ω	16.1	12.6	10.0	8.0	5.2
6	Indoor unit sensor for heat-exchanger temp.	DTN-C103J40	Maximum input 34mA	°C	-12	0	25	50	
				k Ω	62.29	32.82	10.0	3.59	
7	Compressor	YH277JA	Output (Rated) 1.9kW, 2pole, 3phase, 380/415V, 50Hz						
8	Outdoor unit fan motor	AF-230-63P	Output (Rated) 63W, 6pole, 1phase, 230V, 50Hz						
9	Running capacitor for outdoor fan motor	EEP2G405HQA114	AC400V, 4 μ F						
10	Magnetic contactor	ME-20F-FS	AC230V, 50Hz, 3alb						
11	High pressure switch	HTB-T317	Tripping pressure 30kg/cm ² G Resetting pressure 23kg/cm ² G						
12	Solenoid coil	L270021	AC240V, 50/60Hz						
13	Crankcase heater		AC240V, 40W						
14	Sensor for defrosting	DTN-C103J40	Maximum input 15.5mA	°C	-12	10			
				k Ω	67.5	21.3			
15	Overcurrent relay	RC4-20S	Tripping current 7.5A, Resetting auto.						
16	Fuse		1A						
17	Relay	SA106H	AC230V, 50/60Hz						
18	Sensor for cooling operation in low ambient temperature	DTN-C103J40	Maximum input 34mA	°C	-12	0	25	50	
				k Ω	62.29	32.82	10.0	3.59	
19	Inner overload relay	CS-7	Tripping temperature 101°C Resetting temperature 82°C						

4.3 RAV-1003CHE8/S1003HE8

NO.	PARTS NAME	TYPE	SPECIFICATIONS						
1	Indoor unit fan motor	AF-230-100-4P	Output (Rated) 100W, 4pole, 1phase, 230V, 50Hz						
2	Running capacitor for indoor fan motor	40FAFN305UJ	AC400V, 3 μ F						
3	Transformer	FT12	187~264V						
4	Induced draft motor	SGM-220-5A	AC220/240V						
5	Sensor for room temperature		Maximum input 450mW	$^{\circ}$ C k Ω	15 16.1	20 12.6	25 10.0	30 8.0	40 5.2
6	Indoor unit sensor for heat-exchanger temp.	DTN-C103J40	Maximum input 34mA	$^{\circ}$ C k Ω	-12 62.29	0 32.82	25 10.0	50 3.59	
7	Compressor	YH406JA	Output (Rated) 3.0kW, 2pole, 3phase, 380/415V, 50Hz						
8	Outdoor unit fan motor	AF-230-63N AF-230-39N	Output (Rated) 63W, 6pole, 1phase, 230V, 50Hz Output (Rated) 39W, 6pole, 1phase, 230V, 50Hz						
9	Running capacitor for outdoor fan motor	35FAFN405UJ	AC350V, 4 μ F						
10	Magnetic contactor	E-20F-FS	AC230V, 50Hz						
11	High pressure switch	HTB-T317	Tripping pressure 30kg/cm ² G Resetting pressure 23kg/cm ² G						
12	Solenoid coil	L270021	AC240V						
13	Crankcase heater		AC240V, 58W						
14	Sensor for defrosting		Maximum input 15.5mA	$^{\circ}$ C k Ω	-12 67.5	10 21.3			
15	Sensor for cooling operation in low ambient temperature	DTN-C103J40	Maximum input 34mA	$^{\circ}$ C k Ω	-12 62.29	0 32.82	25 10.0	50 3.59	
16	Overcurrent relay	RC4-20RSI	Tripping current 9A, Resetting manual						
17	Fuse		1A						

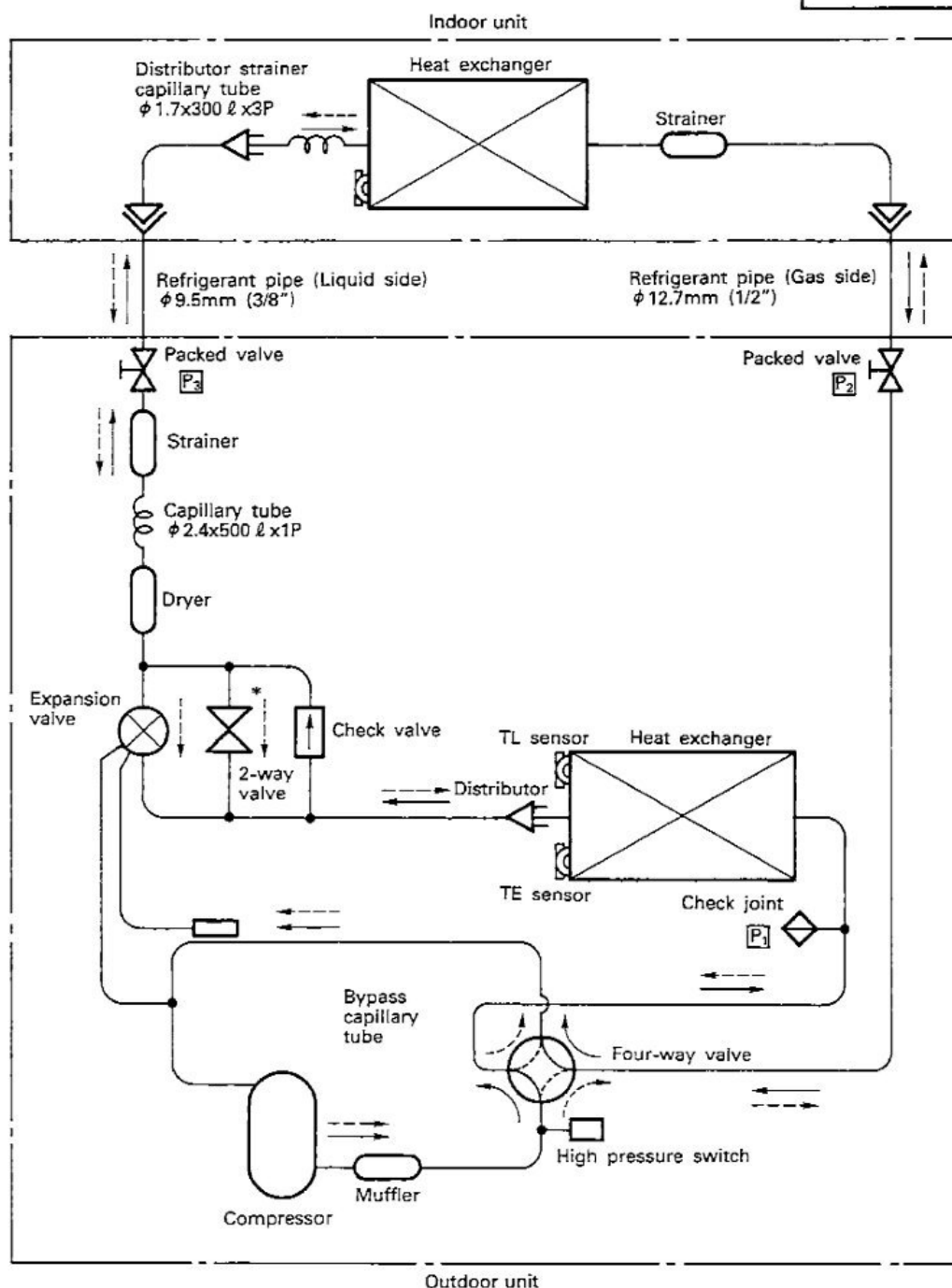
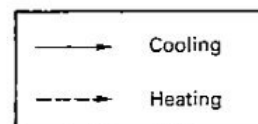
4.4 RAV-1253CHE8/S1253HE8

NO.	PARTS NAME	TYPE	SPECIFICATIONS						
1	Indoor unit fan motor	STF-200-140-4E	Output (Rated) 140W, 4pole, 1phase, 200V, 50/60Hz						
2	Running capacitor for indoor fan motor	EAL40M505UF2	AC400V, 5 μ F						
3	Transformer	FT12	187~264V						
4	Induced draft motor	SGM-220-5A	AC220/240V						
5	Sensor for room temperature		Maximum input 450mW	°C	15	20	25	30	40
				k Ω	16.1	12.6	10.0	8.0	5.2
6	Indoor unit sensor for heat-exchanger temp.	DTN-C103J40	Maximum input 34mA	°C	-12	0	25	50	
				k Ω	62.29	32.82	10.0	3.59	
7	Compressor	YH506JA	Output (Rated) 3.75kW, 2pole, 3phase, 380/415V, 50Hz						
8	Outdoor unit fan motor	AF-230-63N	Output (Rated) 63W, 6pole, 1phase, 230V, 50Hz						
		AF-230-39N	Output (Rated) 39W, 6pole, 1phase, 230V, 50Hz						
9	Running capacitor for outdoor fan motor	35FAFN405UJ	AC350V, 4 μ F						
10	Magnetic contactor	C-25	AC230V, 50Hz						
11	High pressure switch	HTB-T317	Tripping pressure 30kg/cm ² G Resetting pressure 23kg/cm ² G						
12	Solenoid coil	L270021	AC240V						
13	Crankcase heater		AC240V, 58W						
14	Sensor for defrosting		Maximum input 15.5mA	°C	-12	0	25	50	
				k Ω	67.5	21.3			
15	Sensor for cooling operation in low ambient temperature	DTN-C103J40	Maximum input 34mA	°C	-12	0	25	50	
				k Ω	62.29	32.82	10.0	3.59	
16	Overcurrent relay	RC4-35EE1	Tripping current 12A, Resetting manual						
17	Fuse		1A						

5. REFRIGERANT PIPING DIAGRAM

Indoor unit
RAV-453CHE

Outdoor unit
RAV-S453HE



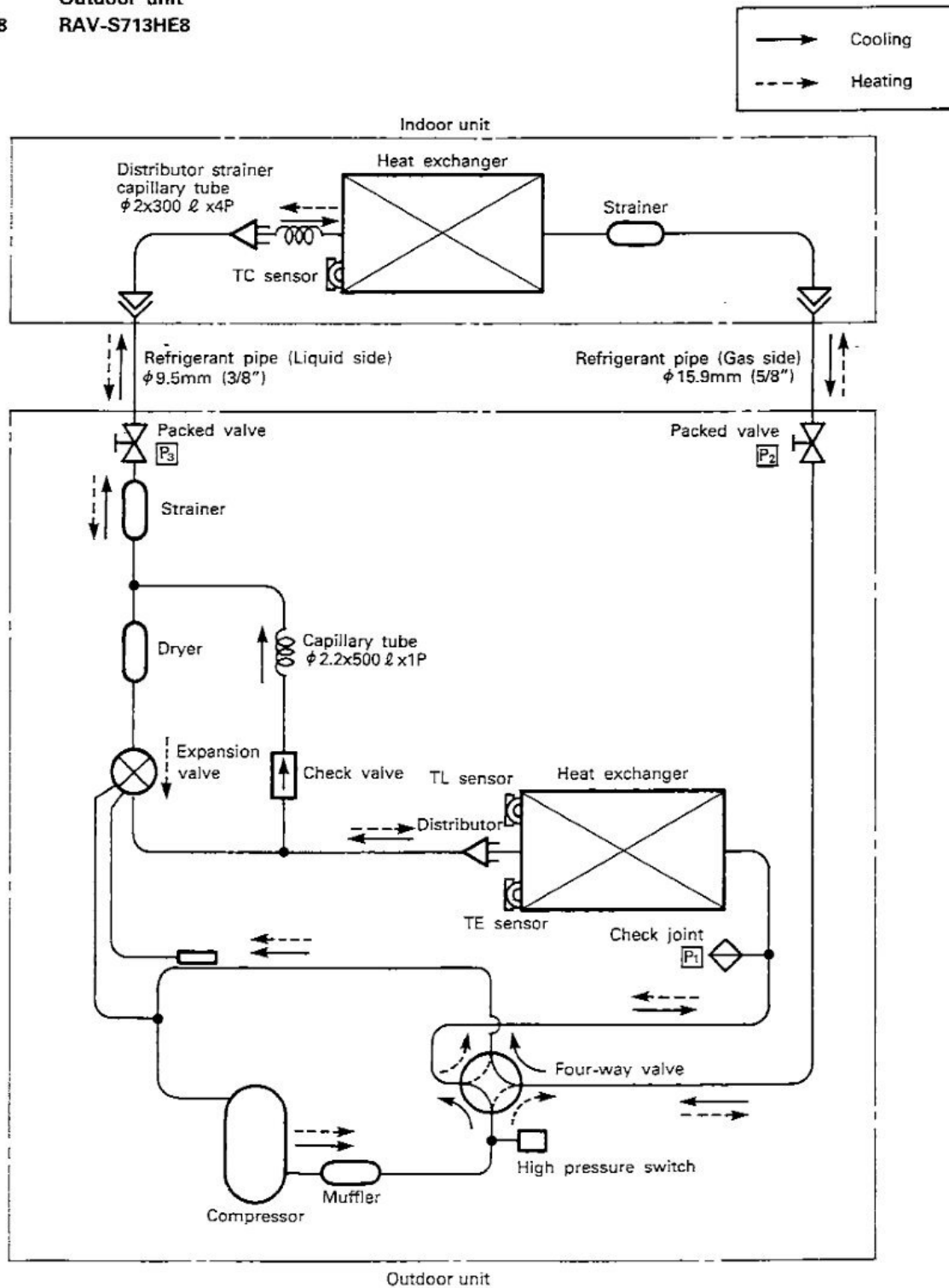
* 2-way valve will open when compressor will go off in heating operation.

Line Pressure

	Cooling	Heating
P_1	High pressure	Low pressure
P_2	Low pressure	High pressure
P_3	Medium pressure	Medium pressure

Indoor unit
IV-713CHE8

Outdoor unit
RAV-S713HE8

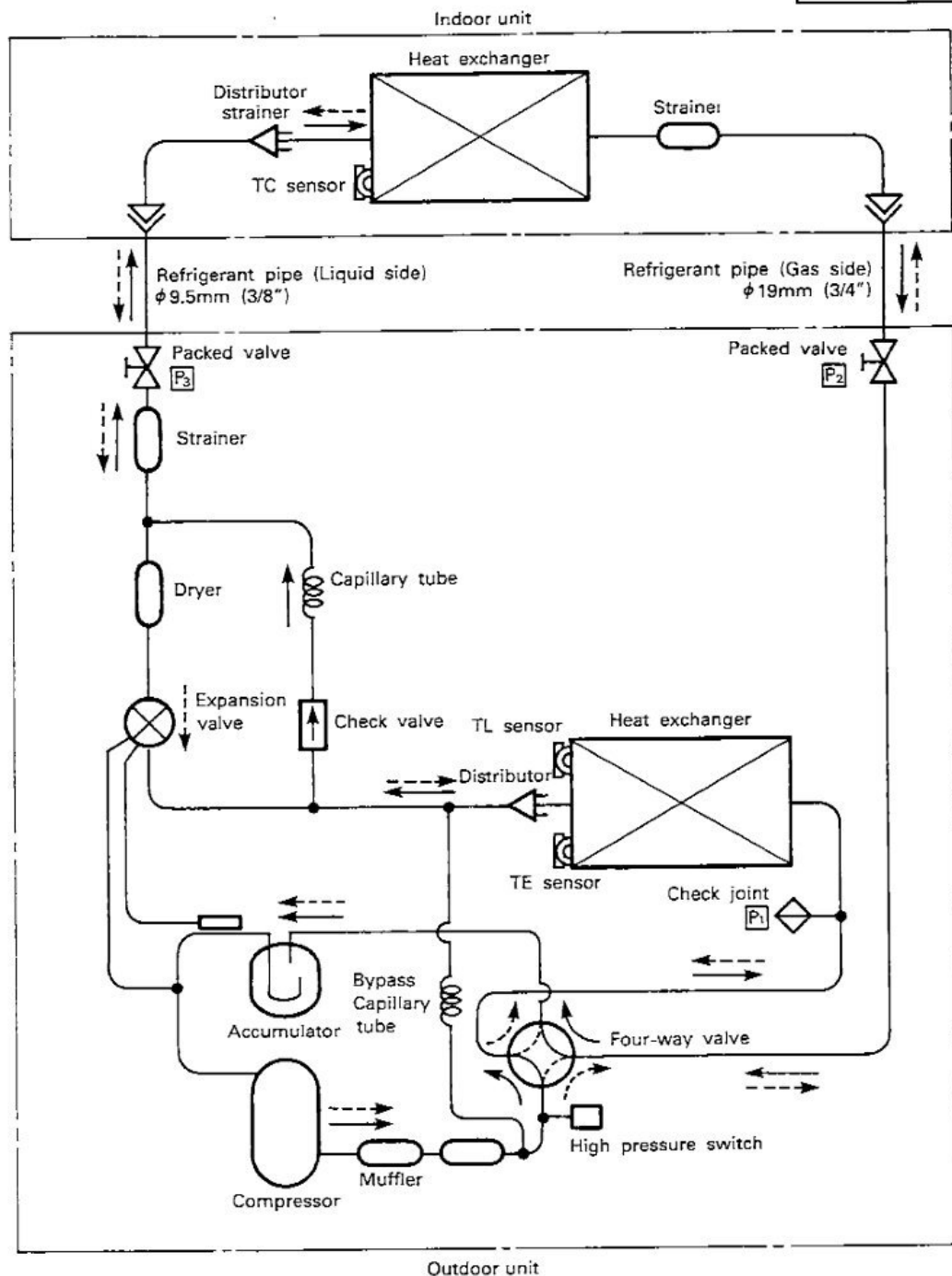
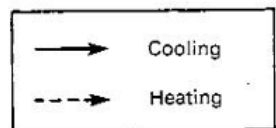


Line Pressure

	Cooling	Heating
P ₁	High pressure	Low pressure
P ₂	Low pressure	High pressure
P ₃	Medium pressure	Medium pressure

Indoor unit
RAV-1003CHE8
RAV-1253CHE8

Outdoor unit
RAV-S1003HE8
RAV-S1253HE8

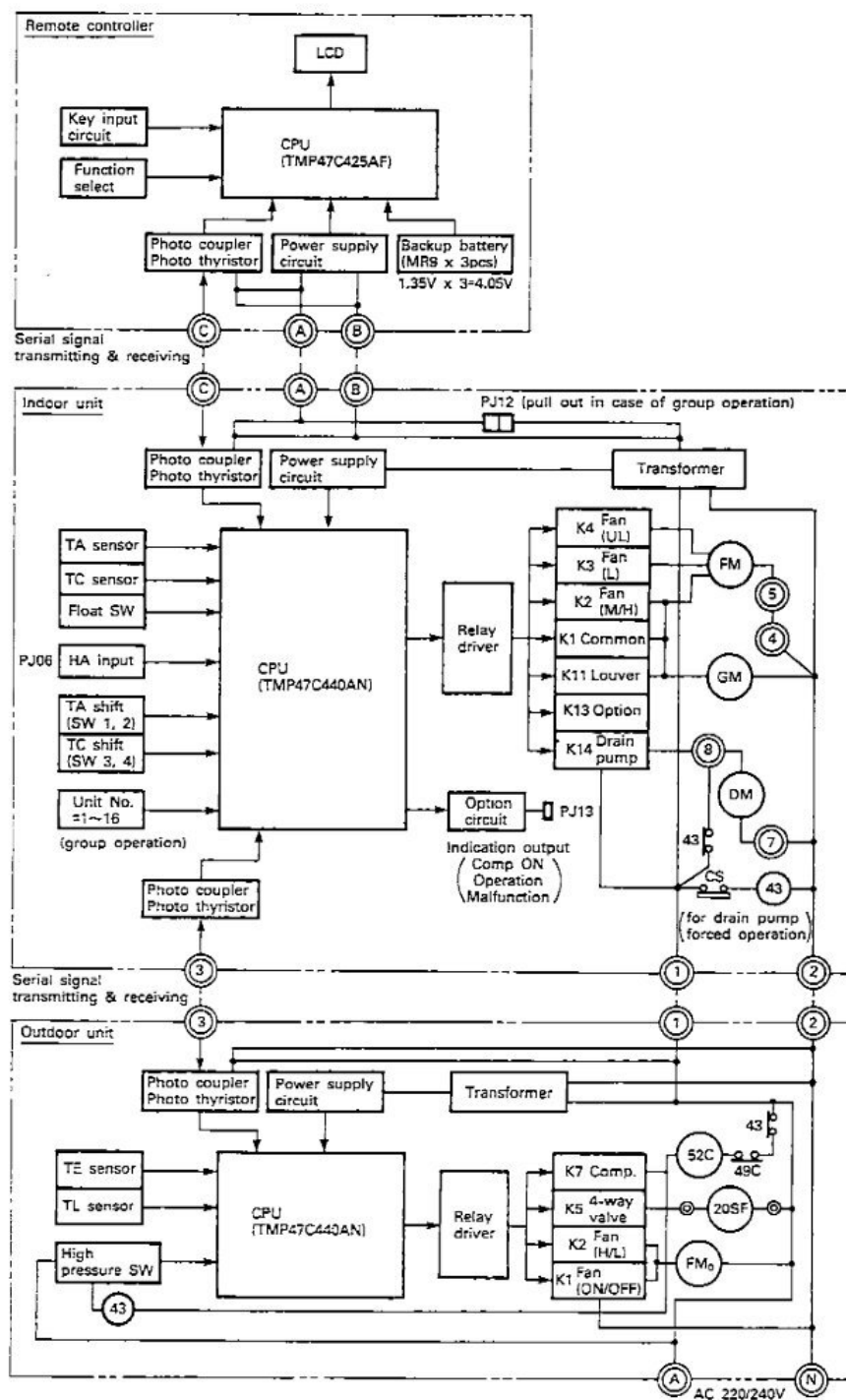


Model	Capillary tube	Bypass Capillary tube
RAV-S1003HE8	ID φ 2.4 X 650 ℓ X 1P	ID φ 1.7 X 1,000 ℓ X 1P
RAV-S1253HE8	ID φ 3 X 600 ℓ X 1P	ID φ 2 X 1,000 ℓ X 1P
RAV-1003CHE8	ID φ 2 X 300 ℓ X 4P	
RAV-1253CHE8	ID φ 2 X 300 ℓ X 8P	

Line Pressure

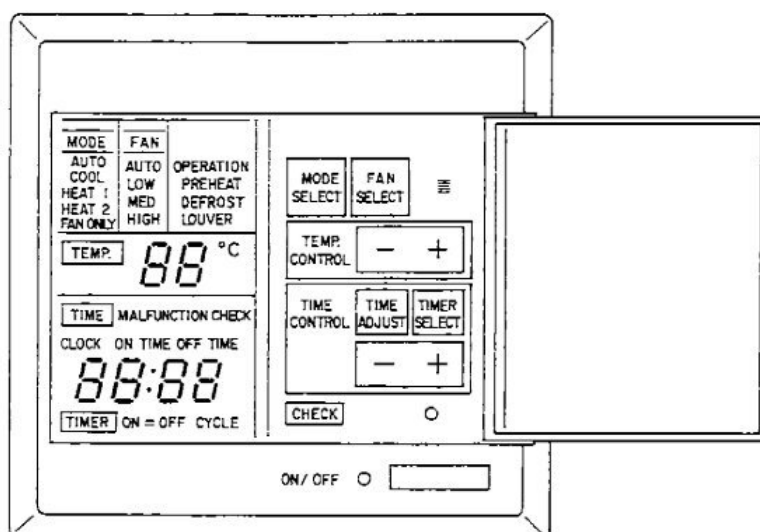
	Cooling	Heating
P ₁	High pressure	Low pressure
P ₂	Low pressure	High pressure
P ₃	Medium pressure	Medium pressure

6. CONTROL CIRCUIT BLOCK DIAGRAM



7. REMOTE CONTROLLER

7.1 Remote controller



BUTTON	INDICATOR		OPERATION
ON/OFF		LED (RED)	Run/Stop
MODE SELECT	<u>MODE</u>	AUTO COOL HEAT 1 HEAT 2 FAN ONLY	Auto Changeover Cooling Heating Heating (with indoor fan operation at defrost) Fan only
FAN SELECT	<u>FAN</u>	AUTO LOW MED HIGH	Auto Fan Speed Control Low Fan Speed Med. Fan Speed High Fan Speed
— — — —		Louver	Auto Louver
TEMP. CONTROL	<u>TEMP.</u>	88 °C	Temperature setting
TIME ADJUST	<u>TIME</u>	(1) CLOCK (2) ON TIME (3) OFF TIME 88:88	(1) Present Time Adjust (2) ON Time Setting (3) OFF Time Setting
TIMER SELECT	<u>TIMER</u>	ON OFF ON → OFF ON ← OFF CYCLE	ON Timer OFF Timer ON → OFF Timer OFF ← ON Timer 24H Cycle Timer

7.3 Timer operation

Continuous operation and timer operations are available. The setting of timer operation can be done as follows:
ON, OFF, ON → OFF, OFF → ON, ON ← → OFF CYCLE.

3-1) Time display

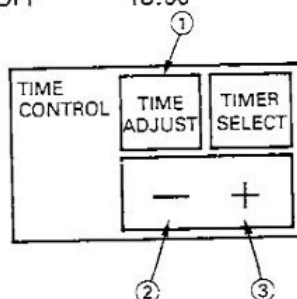
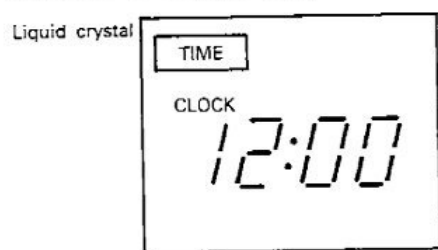
The present time is always displayed

The display of the ON/OFF time is only in setting the time.

Once set, it will not changed even after carrying out the timer operation until the timer is reset.

Initial set time	The present time	12:00
	The time of ON	6:00
	The time of OFF	18:00

3-2) How to set the time



As to (-) and (+), change takes place by one minute by pressing once and 10 min./0.25 sec. by pressing continuously.

How to set the present time



[TIME ADJUST] switch is pressed.
[CLOCK] and Time figures flash.

- 2) Time is set by Time setting switch (-) or (+). The setting is finished when releasing. Pressing [TIME ADJUST] three times gives the display of the present time.
(If left as it is, after 15 sec. the display will go back to the present time).

How to set ON TIME



[TIME ADJUST] switch is pressed twice.
[ON TIME] and Time figures flash.

- 2) Time is set by Time setting switch (-) or (+). The setting is finished when releasing. Pressing [TIME ADJUST] twice gives the display of the present time.
(If left as it is, after 15 sec. the display will go back to the present time)

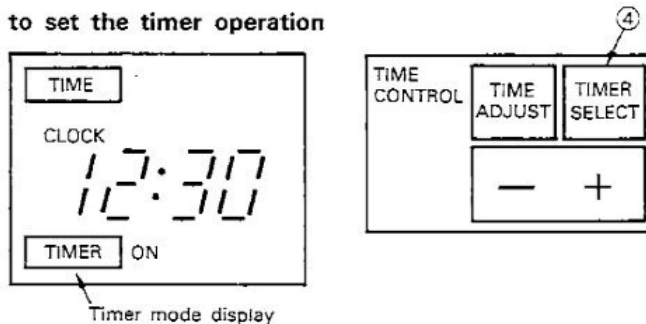
How to set OFF TIME



[TIME ADJUST] switch is pressed three times.
[OFF TIME] and Time figures flash.

- 2) Time is set by Time setting switch (-) or (+). The setting is finished when releasing. Pressing [TIME ADJUST] once gives the display of the present time.
(If left as it is, after 15 sec. the display will go back to the present time)

3-3) How to set the timer operation



The following can be chosen sequentially by pressing [TIMER SELECT] switch:

- 1) [TIMER] ON
- 2) [TIMER] OFF
- 3) [TIMER] ON → OFF
- 4) [TIMER] ON ↔ OFF
- 5) [TIMER] CYCLE

* Be sure to set the present time.

* In case of reoperating after finishing timer operation, if [TIMER SELECT] is not altered, the timer operation will be performed again.

Timer ON operation

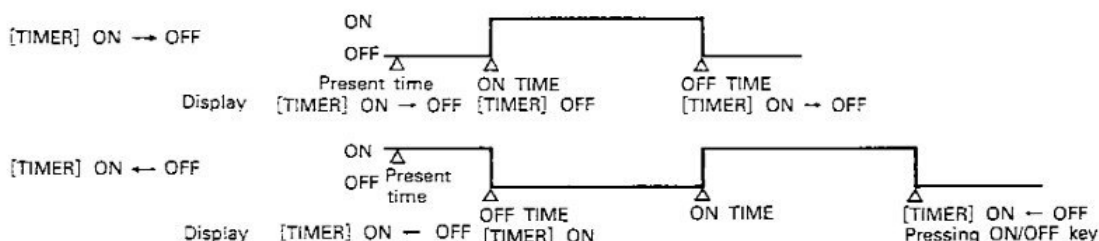
- 1) [TIMER] ON is applied.
- 2) ON/OFF key is pressed. Then LED is lighted.
When the set [ON TIME] comes, the operation starts and OPERATION display comes on the liquid crystal, and the [TIMER] ON display goes off.
- 3) LED and OPERATION display goes off upon pressing ON/OFF key for stopping and [TIMER] ON is displayed.

Timer OFF operation

- 1) [TIMER] OFF is applied.
- 2) ON/OFF key is pressed. Then LED is lighted and the operation starts with OPERATION displayed on the liquid crystal.
- 3) When the set [OFF TIME] comes, the operation stops and the LED, OPERATION display goes off with [TIMER] OFF displayed.

ON ↔ OFF timer operation

- 1) [TIMER] ON → OFF or [TIMER] ON ↔ OFF is applied.
- 2) ON/OFF key is pressed. LED comes on and the operation is performed as below:

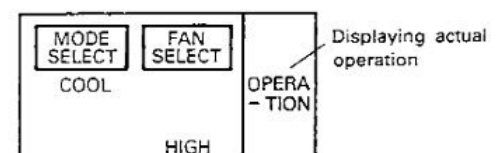


Repeated operation

- 1) [TIMER] CYCLE is applied.
- 2) ON/OFF key is pressed. Then LED is lighted and ON ↔ OFF timer operation is repeated according to the ON time and OFF time (repeating every day as it is a 24-hour timer).
- 3) The operation key is pressed. LED goes off and operation stops.

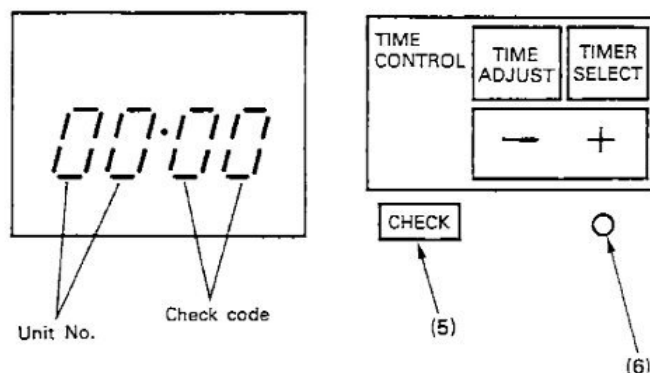
Timer stand-by display and operation display

Waiting on the timer is displayed by LED lighting while the actual operation is displayed on OPERATION on liquid crystal.



7.4 Malfunction check monitor

4-1) The times of thermostat ON as well as the check code are displayed on the time display area by pressing CHECK key.



(5) Check key

Provides check code display by pressing for one second and indoor microcomputer reset by pressing for 5 seconds.

- * Remote controller clear by pressing the key for 10 sec. Check code is cleared (normally not used).
- (6) Reset key (pushed by a needle and the like) Resetting remote controller (to the initial setting)

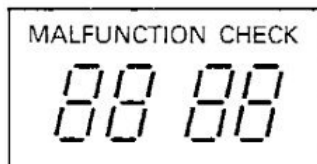
Judgement from operation status

	OPERATION STATUS	CODE	CAUSE
1.	Thermostat stays off in cooling: it is not turned off in heating	0C	Open-circuit in room temperature sensor.
	Thermostat stays off in heating: it is not turned off in cooling		Short-circuit in room temperature sensor.
2.	Indoor fan stays off in heating.	0d	Open-circuit in indoor heat-exchanger sensor.
	Outdoor fan continues ON-OFF operation in heating.		Short-circuit in indoor heat-exchanger sensor.
3.	While indoor unit is in operation, outdoor unit keeps on stoppage Neon lamp comes on.	0b	Abnormality in drain system. Fault of drain pump. Drain pipe clogged.
4.	Though indoor unit operates, outdoor unit keeps on stopping.	04	Abnormality in connecting cable between indoor and outdoor units.
5.	Indoor fan does not work in heating operation. Warm air comes out in cooling operation.	08	4-way valve coil burnt out, pipe clogged, abnormality in indoor heat-exchanger sensor.
6.	Indoor fan at LOW speed in cooling operation with the outdoor remaining in stoppage.	09	Refrigerant gas in shortage. Abnormality in indoor heat-exchanger sensor.
7.	Full stop	18	Open or short-circuit in outdoor TE sensor.
8.	Full stop	19	Open or short-circuit in outdoor TL sensor.
9.	Full stop	21	Pressure switch does not reset within the set time.
10.	Full stop	1C	Pressure switch, overcurrent sensor operated.

4-2) How to read malfunction check monitor display

By pressing [CHECK] key, times of No. 1 unit compressor-ON actuations as well as the check code information of 2 faults x 16 units are displayed on the time display area. (2 sec. per one phenomenon)

<Times of compressor-ON>

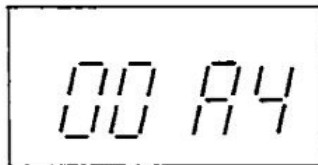


Display in 4 digits of hexadecimal notation

Display in 7 segments

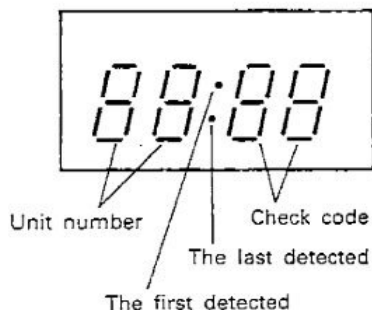


Ex. In case of the number of times of compressor actuations of 164.

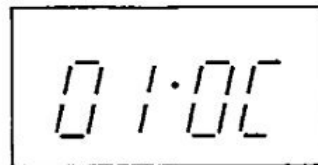


$$16^3 \times 0 + 16^2 \times 0 + 16 \times 10 + 4 = 164$$

<Check code information>

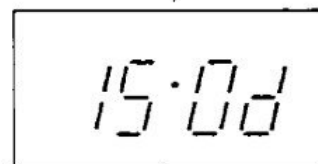
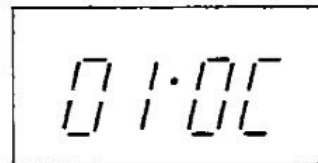


Ex. In case of room temperature sensor of No.1 unit in trouble.



For No. 15 unit, firstly room temperature sensor and secondly connecting cable between the indoor and outdoor units (serial signal) are in trouble.

No display is made if there is no fault.



4-3) List of Check Code

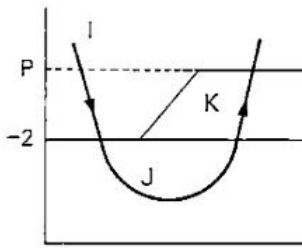
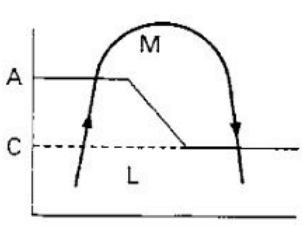
DIAGNOSTIC FUNCTIONS			JUDGEMENT AND ACTION
CHECK CODE	SYMPTOM	STATUS OF AIR CONDITIONER	
01	ROOM TEMP. SENSOR (TA). Out of place, break, short-circuit.	Operation continuing	1. Check for indoor temp. sensor. 2. Check for indoor PC board.
02	INDOOR HEAT-EXCHANGER SENSOR (TC). Out of place, break, short-circuit.	Operation continuing	1. Check for indoor heat-exchanger sensor. 2. Check for indoor PC board.
06	FLOAT SWITCH float circuit out of position, break.	Outdoor unit stops	1. Fault in drain pump. 2. Drain pipe clogged. 3. Check for indoor PC board.
04	RETURN SIGNAL NOT COMING TO INDOOR 1) Wrong wiring in connecting cable (serial signal).	Operation continuing	1. If outdoor unit does not work at all. (1) Check for connecting cable correct wiring. (2) Check for outdoor PC board. 2. If operates normally. Between indoor terminal plates 2 and 3, return signal is : Available: Check for indoor PC board. Not available: Check for outdoor PC board.
08	4-WAY VALVE SYSTEM 1) Indoor heat-exchanger temperature rises, after starting cooling operation. 2) Indoor heat-exchanger temperature drops after starting heating operation.	Operation continuing	1. Check for 4-way valve. 2. Check for 2-way valve and check valve. 3. Wrong with indoor heat exchanger sensor. 4. Check for indoor PC board.
09	OTHER CYCLE SYSTEM 1) Indoor heat exchange temperature does change after starting cooling/heating operation.	Operation continuing	1. Compressor case thermostat, IOL operation. (contactor OFF, compressor stops) 2. Indoor heat-exchange sensor out of place. 3. Check for indoor PC board.
	2) When transmitting instruction for stopping compressor by freeze preventing control.	Outdoor unit stops (indoor fan L)	1. Check for charged amount of refrigerant gas. (Gas shortage → gas supplement, check for gas leaks) 2. Indoor fan locked.

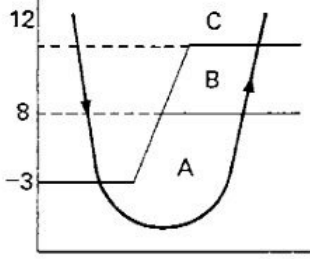
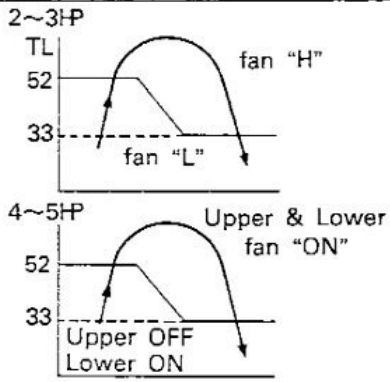
18	DEFROST SENSOR (TE) Out of place, break, short-circuit.	Full stop	1. Check for defrosting sensor. 2. Check for outdoor PC board.
19	OUTDOOR HEAT EXCHANGER SENSOR (TL) Out of place, break, short-circuit.	Full stop	1. Check for outdoor heat exchange sensor. 2. Check for outdoor PC board.
21	HIGH PRESSURE SWITCH High pressure switch does not reset. (5 sec : in cooling) (30 sec : in heating)	Full stop	1. Check for high pressure switch. 2. Check for outdoor PC board.
11	OTHER ABNORMALITY OF OUTDOOR UNIT Compressor does not operate. Start once, but soon after stop by OCR	Full stop	1. Check for compressor. 2. Check for wiring of compressor. (lack of phase, short circuit) 3. Check for voltage. 4. Check for outdoor PC board.

8. OUTLINE OF CONTROL CIRCUIT

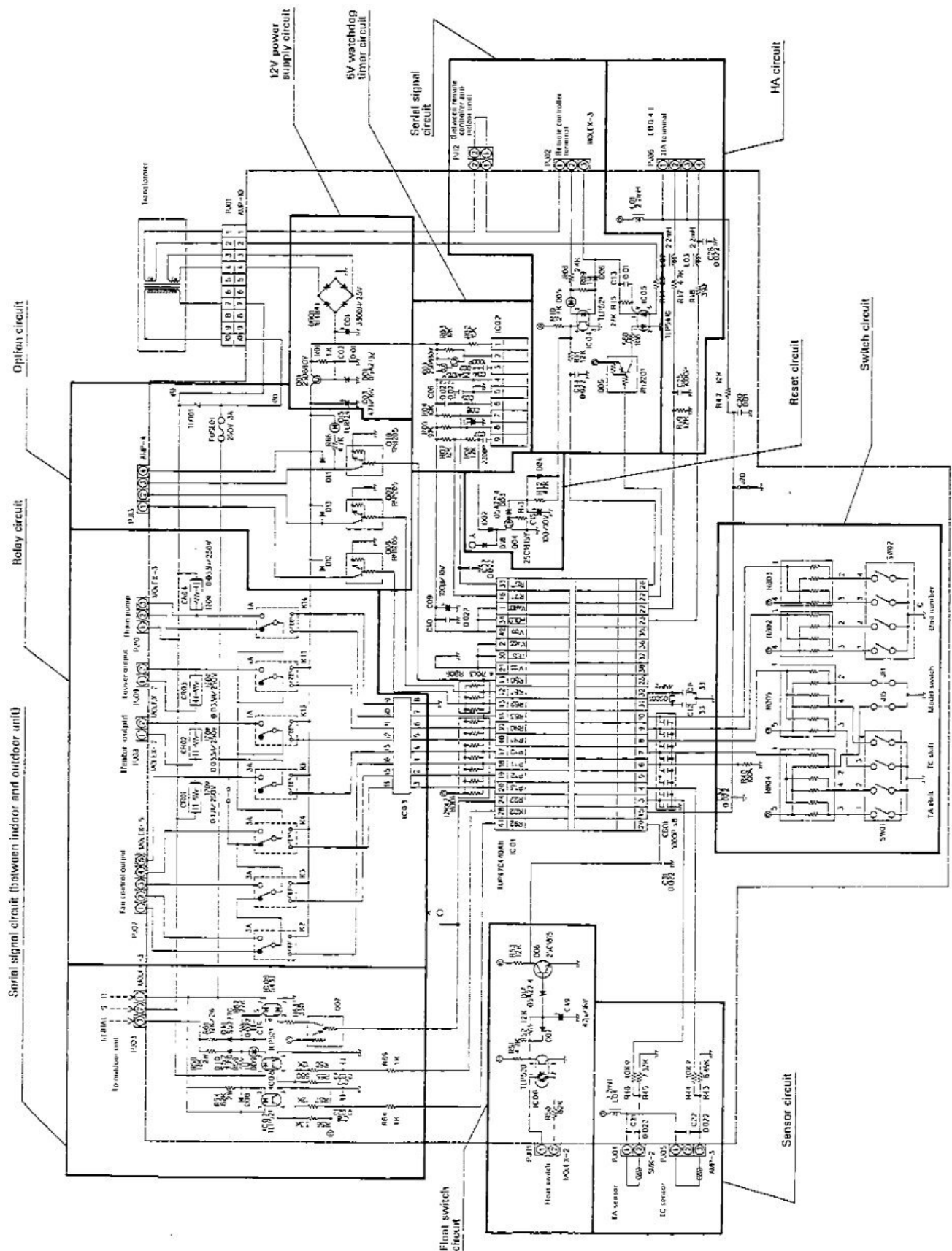
NO.	ITEM	OUTLINE OF SPECIFICATIONS	REMARKS																		
1	Discrimination	Discrimination of outdoor unit is performed either in the reset of power source or when stopping from operating condition, and the controlling is changed over in accordance with the result of discrimination. [Heating S1] is used as the discriminating signal, [Inverter heat pump], [Normal heat pump], and [Cooling only] being discriminated according to the contents of reverse signal from the outdoor unit.																			
2	Operation change-over	Operation mode is changed over according to operation mode select instruction from the remote controller. <table><tr><th>REMOTE CONTROLLER INSTRUCTION</th><th>OUTLINE OF CONTROL</th></tr><tr><td>Stop</td><td>Stopping air conditioner</td></tr><tr><td>Auto</td><td>Performing automatic change-over</td></tr><tr><td>Cool</td><td>Performing cooling operation</td></tr><tr><td>Heat 1</td><td>Performing heating operation</td></tr><tr><td>Heat 2</td><td>Performing heating operation with indoor fan operation at defrosting</td></tr><tr><td>Fan only</td><td>Performing fan only operation</td></tr></table>	REMOTE CONTROLLER INSTRUCTION	OUTLINE OF CONTROL	Stop	Stopping air conditioner	Auto	Performing automatic change-over	Cool	Performing cooling operation	Heat 1	Performing heating operation	Heat 2	Performing heating operation with indoor fan operation at defrosting	Fan only	Performing fan only operation					
REMOTE CONTROLLER INSTRUCTION	OUTLINE OF CONTROL																				
Stop	Stopping air conditioner																				
Auto	Performing automatic change-over																				
Cool	Performing cooling operation																				
Heat 1	Performing heating operation																				
Heat 2	Performing heating operation with indoor fan operation at defrosting																				
Fan only	Performing fan only operation																				
3	Controlling room temperature	3-1 Adjusting range (°C) <table><tr><td></td><td>In cooling</td><td>In heating</td></tr><tr><td>Remote controller setting temperature</td><td>18 ~ 29</td><td>18 ~ 29</td></tr><tr><td>Operating temperature</td><td>18 ~ 29</td><td>20 ~ 31</td></tr></table> 3-2 Operating point is compressor - off. 3-3 Operating temperature accuracy: ±1°C 3-4 Differential: 1 deg		In cooling	In heating	Remote controller setting temperature	18 ~ 29	18 ~ 29	Operating temperature	18 ~ 29	20 ~ 31										
	In cooling	In heating																			
Remote controller setting temperature	18 ~ 29	18 ~ 29																			
Operating temperature	18 ~ 29	20 ~ 31																			
	Correcting temperature compensation	3-5 Room temperature controlled in the heating operation can be corrected by dip switch of indoor microcomputer. <table><tr><td>Dip switch</td><td>1</td><td>ON</td><td>ON</td><td>OFF</td><td>OFF</td></tr><tr><td>Setting</td><td>2</td><td>ON</td><td>OFF</td><td>ON</td><td>OFF</td></tr><tr><td>Control temperature compensation</td><td></td><td>0deg</td><td>2deg</td><td>4deg</td><td>6deg</td></tr></table>	Dip switch	1	ON	ON	OFF	OFF	Setting	2	ON	OFF	ON	OFF	Control temperature compensation		0deg	2deg	4deg	6deg	Ts(Max)=35°C
Dip switch	1	ON	ON	OFF	OFF																
Setting	2	ON	OFF	ON	OFF																
Control temperature compensation		0deg	2deg	4deg	6deg																
4	Automatic capacity control (Thermo-control)	4-1 Instruction of operation frequency is given to outdoor unit based on the table below according to the difference between Ta and Ts. 4-2 Compressor is turned on at the frequency instruction of [S3] or higher, and turned off below [S3].																			

NO.	ITEM	OUTLINE OF SPECIFICATIONS	REMARKS
		- Cooling - - Heating - - Auto-changeover -	[Stop] is cold draft prevention by T_c.
5	Fan speed control	5-1 [HIGH], [MED.], [LOW] and [AUTO] are available. 5-2 [Ultra low] or [Stop] when thermostat is turned off while heating is being performed. 5-3 In the auto fan, the fan speed is changed by the difference between T_a and T_s, as shown below.	
6	Cold draft preventing control	When performing heating operation, indoor fan control is carried out as follows based on temperature detection of T_c sensor. C zone: Depending upon fan speed setting of the remote controller B zone: Indoor fan at "L" A zone: Fan stop	

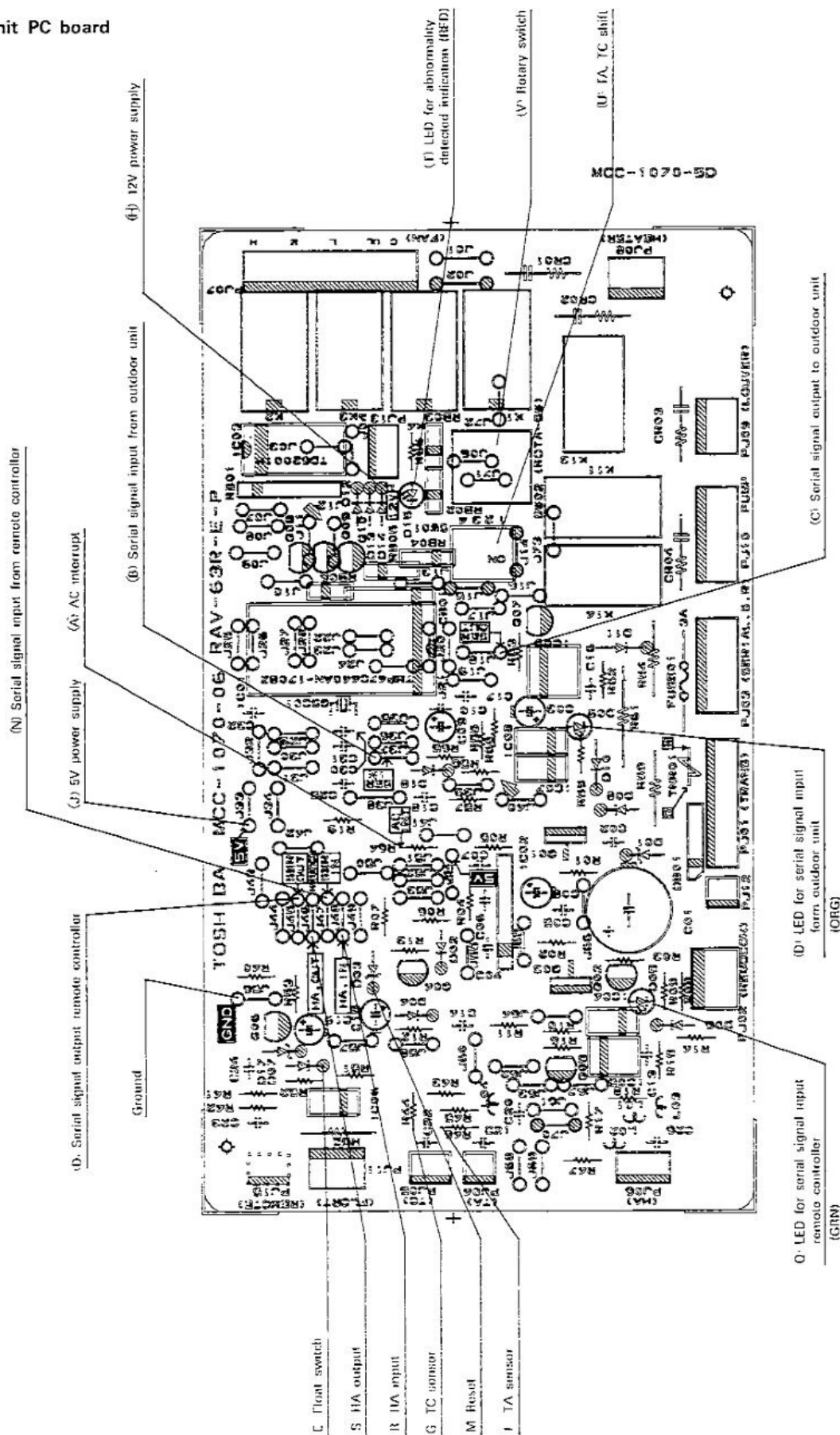
NO.	ITEM	OUTLINE OF SPECIFICATIONS	REMARKS																		
7	Freeze preventing control (Low temp. release)	When performing cooling operation, the following control is done based on temperature detection of Tc sensor. ① When starting the operation, the point P is made +3°C. ② Frequency instructed to the outdoor unit is reduced when [J] zone is detected for 1 minute continuously. ③ Frequency instructed to the outdoor unit is reduced each time 2 minutes have elapsed thereafter. ④ When [K] zone is detected, timer counting is discontinued and held on. ⑤ When [I] zone is detected, timer is cleared for returning back to ordinary operation. ⑥ When frequency instructed to the outdoor unit has become H, the point P is changed to -12°C to be covered by check display. When [I] zone is reached, the temperature is returned back to +3°C.	 H=OFF																		
8	High temperature release control	When performing heating operation, the following control is done based on temperature detection of Tc sensor. ① In [M] zone, release signal is transmitted. Outdoor fan is turned off at the shortest for 3 minutes based on this signal. ② The control point for A and C can be chosen from the below table: <table border="1" data-bbox="544 1328 1248 1482"> <tr> <td>Dip switch</td><td>3</td><td>ON</td><td>ON</td><td>OFF</td><td>OFF</td></tr> <tr> <td>Setting</td><td>4</td><td>ON</td><td>OFF</td><td>ON</td><td>OFF</td></tr> <tr> <td>A/C (°C)</td><td></td><td>54/52</td><td>58/56</td><td>60/58</td><td>-</td></tr> </table>	Dip switch	3	ON	ON	OFF	OFF	Setting	4	ON	OFF	ON	OFF	A/C (°C)		54/52	58/56	60/58	-	 Interval operation of outdoor upper fan at low is done in the outdoor unit select B mode. (Outdoor fan) B=4HP, 5HP
Dip switch	3	ON	ON	OFF	OFF																
Setting	4	ON	OFF	ON	OFF																
A/C (°C)		54/52	58/56	60/58	-																
9	Residual heat removal	When stoppage takes place in [HEAT 2] operation, indoor fan is operated in [LOW] for 30 sec.																			
10	Auto louver control	When receiving louver signal from remote controller, auto louver operation is performed if indoor fan is being in operation.	Provided for ceiling type																		
11	HA control	The air conditioner can be turned on and off by pulse signal DC5V.	External signal																		
12	Test operation	12-1 If Remote controller's ON/OFF switch is pressed 5 seconds continuously, the unit goes into test run mode, and fixed-frequency operation is performed with the indoor fan in the [HIGH]. 12-2 After continuing the operation for 30 minutes, [Fan only] operation is initiated.	Instructed frequency [S7]																		

NO.	ITEM	OUTLINE OF SPECIFICATIONS	REMARKS
13	High pressure release	The following control is performed when high pressure switch of the outdoor unit is actuated. ① In cooling operation Compressor is turned off and if the high pressure switch does not reset for 5 seconds continuously thereafter, it is judged abnormal. ② In heating operation Compressor is turned off and if the high pressure switch does not reset for 30 seconds continuously thereafter, it is judged abnormal. If the switch resets within 30 sec., the compressor restarts 2 minutes and 30 sec. later. And if this process is repeated, the release by outdoor fan and shift of compressor-on point will be done. ③ In defrosting operation Compressor is turned off, the operation returning back to heating operation.	<Outdoor unit control> LED lamp comes on in abnormal condition, being abnormal code transmitted to indoor unit.
14	Defrosting	14-1 In heating operation, defrosting is made based on outdoor heat exchange temperature T_e. 14-2 When cumulative working time of the compressor in [A] zone has amounted to 55 minutes, defrosting operation starts. (25 minutes initially) 14-3 The longest defrosting time is 12 minutes, 60 sec. in the case of turning into [B] zone, and immediate returning back when [C] zone is reached. 	<Outdoor unit control>
15	Low ambient cooling	15-1 Control on outdoor fan is made to meet with cooling at low outdoor temperature based on outdoor heat exchange temperature T_L. 15-2 Control by outdoor heat exchange temperature T_L is illustrated in the right. 	<Outdoor unit control>
16	Check display	Fault diagnosis is carried out by check for serial signal transmission and reception with outdoor unit as well as the self check by indoor microcomputer. And check code is transmitted to protective operation/remote controller based on the contents of it. In the remote controller, check code is displayed on the liquid crystal by pressing [CHECK] key.	See other item: Using [TIME] display  Unit No. check code.
17	Anti-restart timer	The outdoor unit delays restarting for 2.5 min. to prevent short cyclic compressor operation.	
18	Group operation control	Up to 16 units can be controlled in same setting condition by one remote controller. However, thermo-control function is independent. Respective delayed start time for preventing simultaneous large starting current can be by different setting of the unit No. switch on the indoor PC board.	Refer to P.50

9.2 Indoor unit control circuit diagram



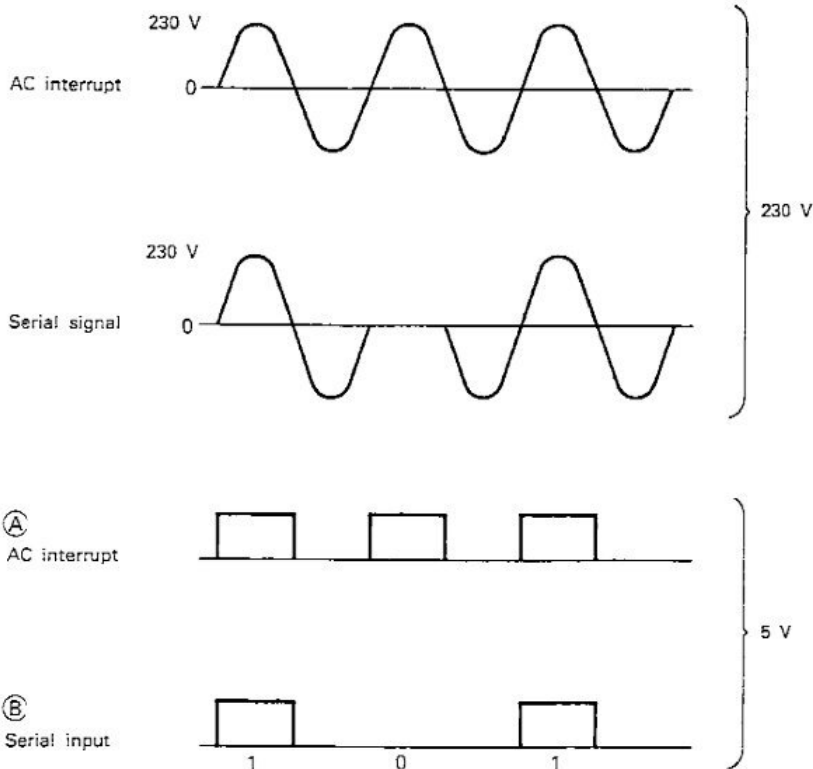
9.3 Indoor unit PC board



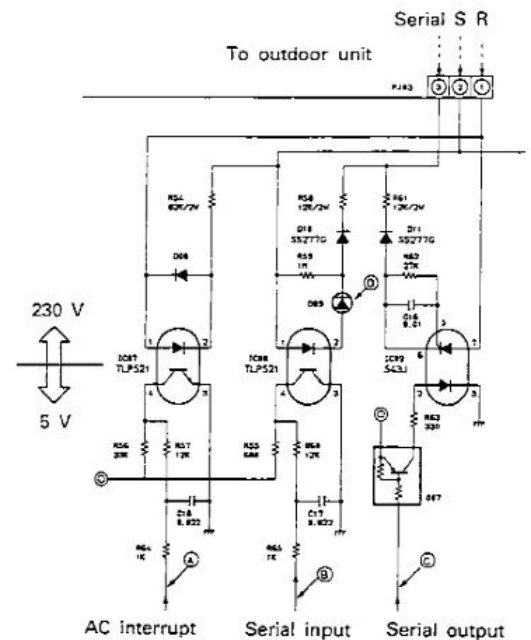
Serial signal circuit (between outdoor and indoor units)

This is a circuit for transmitting and receiving the signals between the indoor and outdoor units in serial signal. As 230V is used for transmitting the signal, the microcomputer section is insulated by means of photo-coupler with the voltage reduced to 5V.

With AC interrupt, judgement is made as to the presence or absence of serial signal based on the reference pulse taken out from the voltage across R and S.

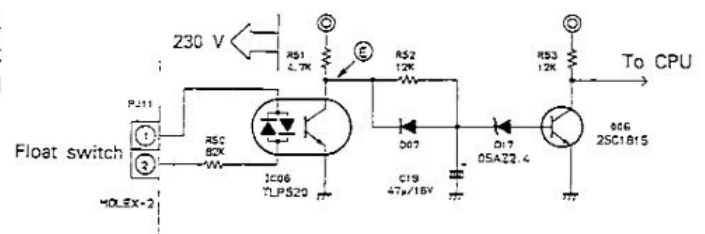


Ⓐ / Ⓑ ... are measurement points on the printed circuit board.
Ⓓ provides flashing (orange) on LED in the serial input.



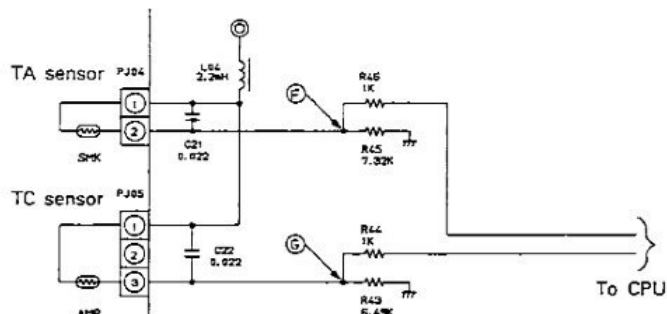
Float switch circuit

In normal condition in which float switch is not operated, 230V is applied across the pins 1 and 2. At this time, point Ⓔ is at the GND level. If the float switch is operated, Ⓔ will be at the level of 5V.



Sensor circuit

This circuit detects the temperature by dividing voltage with resistance and sensor and bringing the voltage value into CPU, using the characteristics of the sensor that resistance varies with different temperatures. TA and TC have the same circuit composition.



When TA and TC are at 25°C approximately, the voltage level is some 2V both at points ⑥ and ⑦.

If ⑥ / ⑦ are at GND or 5V, abnormal condition prevails such as opening or short-circuit of the sensor.

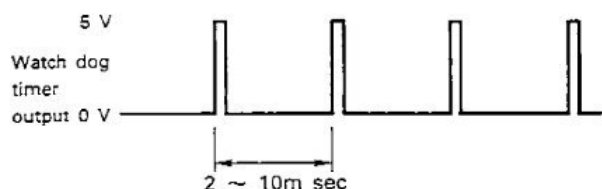
12V power source circuit

Full-wave rectification by diode bridge (DB01) of alternate current supplied from power transformer followed by the provision of transistor (Q01) gives DC12V power source (H).

5V watchdog timer circuit

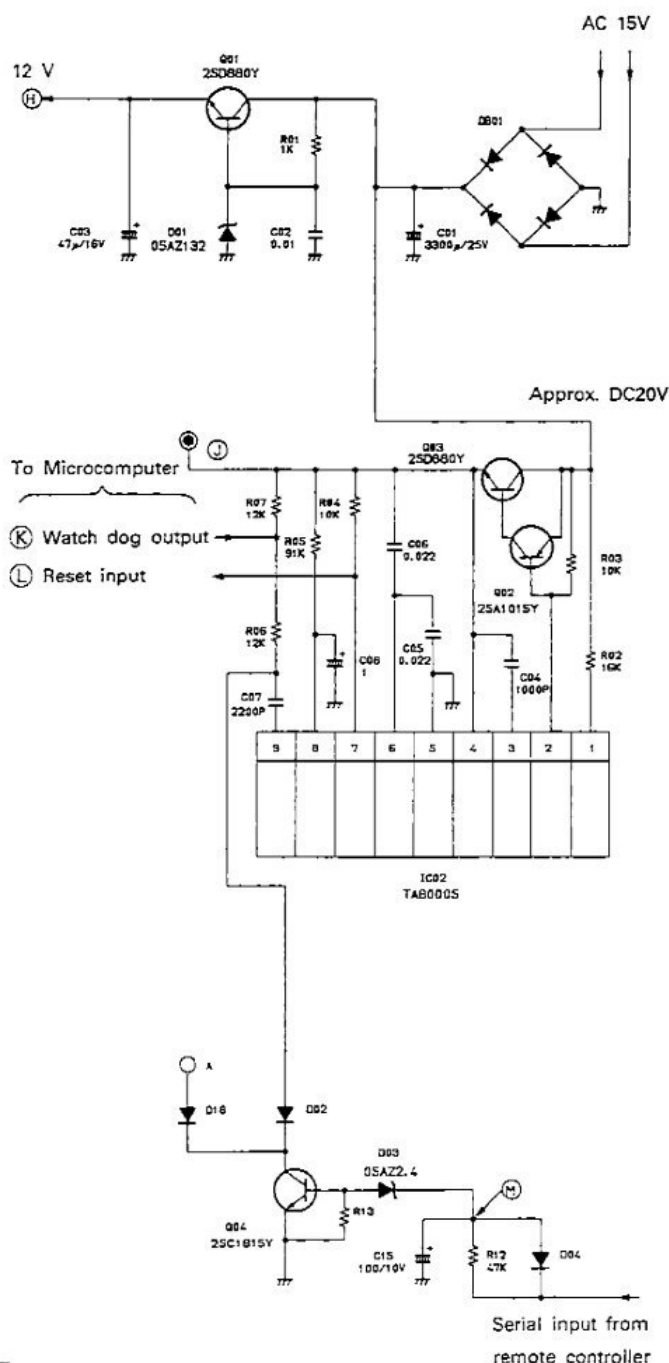
Built-in IC (TA8000S) is used to produce 5V power source (J). Also, it sends signals to reset port (L) of microcomputer which is in stand-by at 0V and starts its operation with the signal of 5V.

Watchdog timer output gives the signal from microcomputer as illustrated below. This indicates that the microcomputer is working in normal routine. For example if the microcomputer is straying due to noise and so on, this waveform is not produced. In case there is no waveform, it plays the role of restoring normal condition by inputting the resetting "0V" to the microcomputer.



Reset circuit

This circuit makes indoor microcomputer reset by way of hardware when you keeps on pressing the check key of remote controller for longer than a predetermined period. It plays the role of resetting microcomputer from the remote controller when it strays. The point (M), which is normally at the level of 5V, drops down to the GND level in the reset operation.

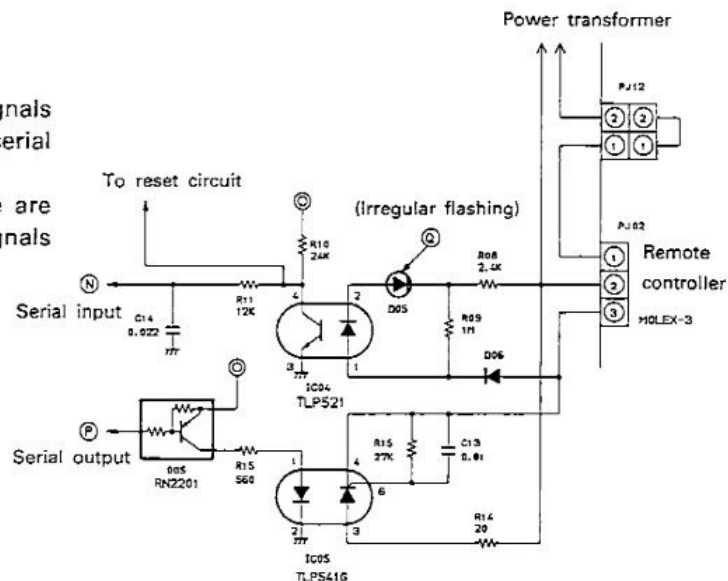
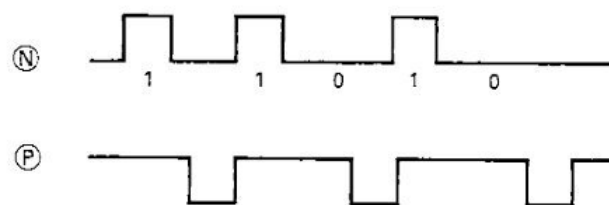


Serial signal circuit

(Between remote controller and indoor unit)

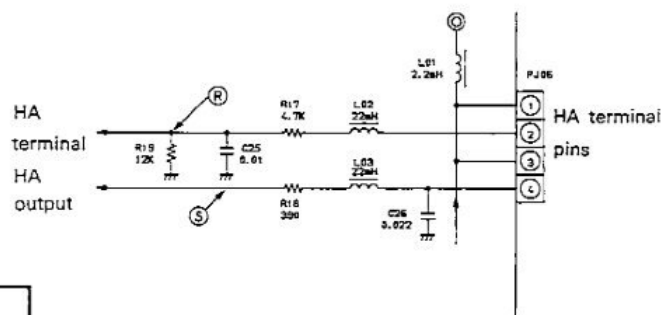
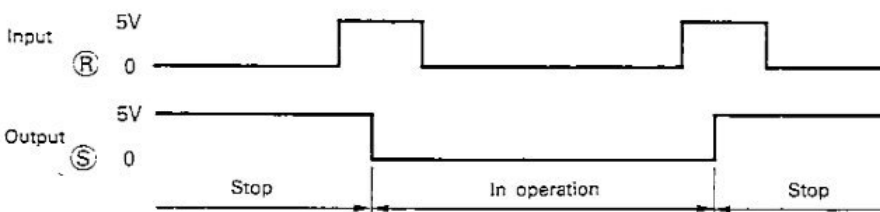
This is the circuit for transmitting and receiving the signals between the remote controller and indoor unit in serial signal.

Point ① is a LED (green) which flashes when there are signals from the remote controller. At ② and ③, the signals as illustrated below are output.



HA circuit

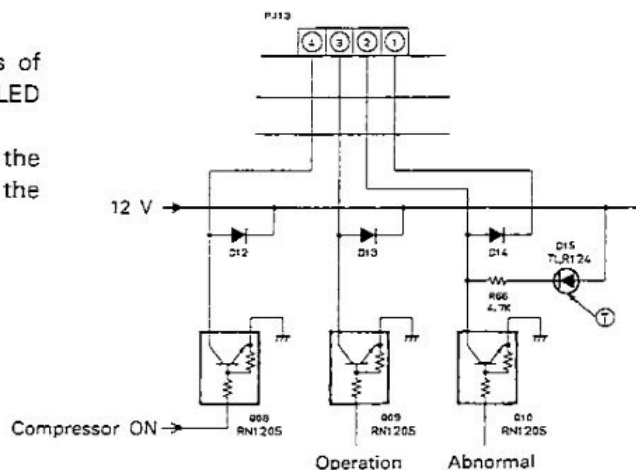
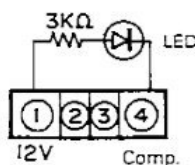
This is a circuit for turning the air conditioner on and off externally. The connector pins 1 and 3 are at 5V. Pin 2, which is signal input, performs on-and-off operation each time 5V signals comes. HA output is at GND in operation and at 5V in stoppage.



Optional circuit

A circuit which allows for the take-out of the signals of abnormal, operation and Compressor-ON. Point ① is a LED which lights at abnormal.

The connector pin 1 outputs 12V. When you want to see the signal of compressor-ON, you can do it simply with the circuit below.

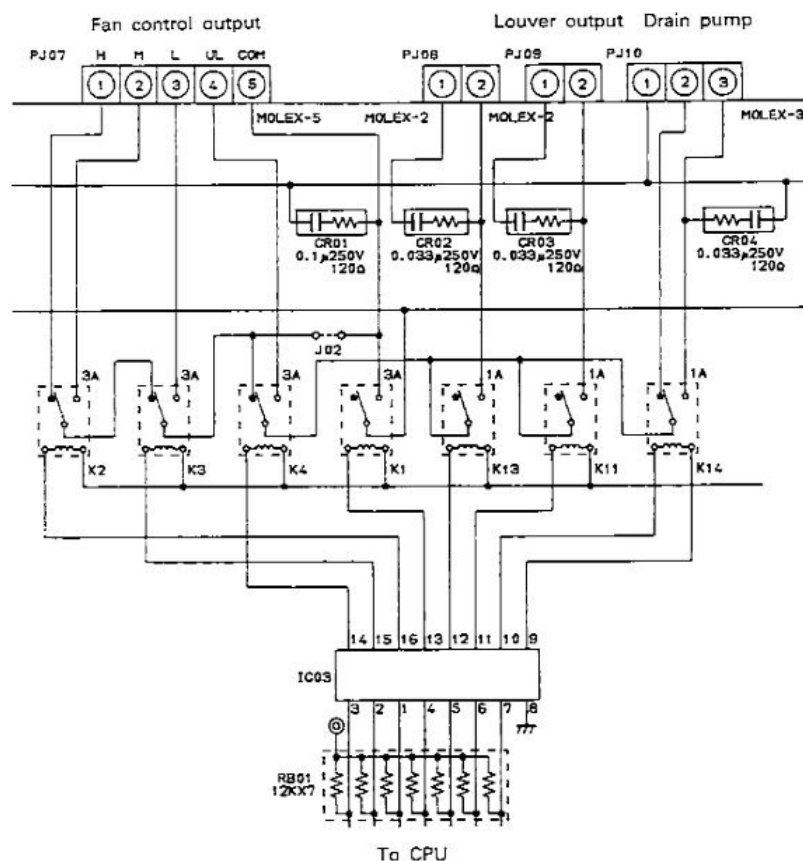


Relay circuit

The relay circuit consists of the diagram in the righthand side.

The relay performs the following functions:

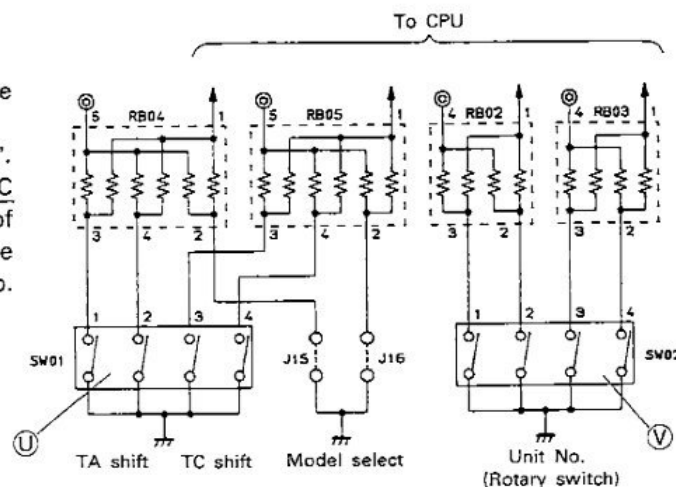
- K1 : Turning fan on and off
 - K2 : Changing over H/M of fan
 - K3 : L tap of fan
 - K4 : UL tap of fan
 - K11: Turning louver on and off
 - K14: Turning drain pump on and off
- (① - ③)



Switch circuit

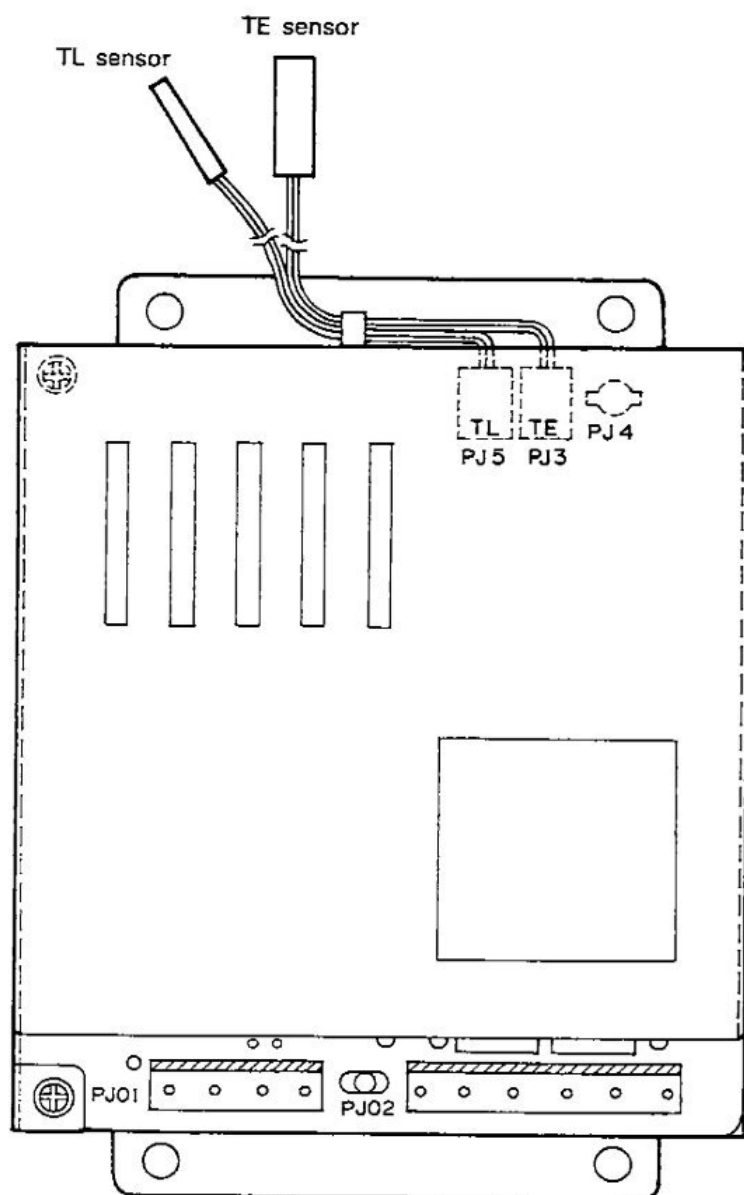
TA shift, TC shift and unit No. are changed over by the switch.

TA shift and TC shift are set in factory with unit No. at "1". In servicing, the setting should be made to the same TA/TC shift as the PC board attached originally. In case of operating one single unit, unit No "1" will do. With the operation of many units (multi units control) the unit No. should be adjusted in such a way as 1, 2, 3.....

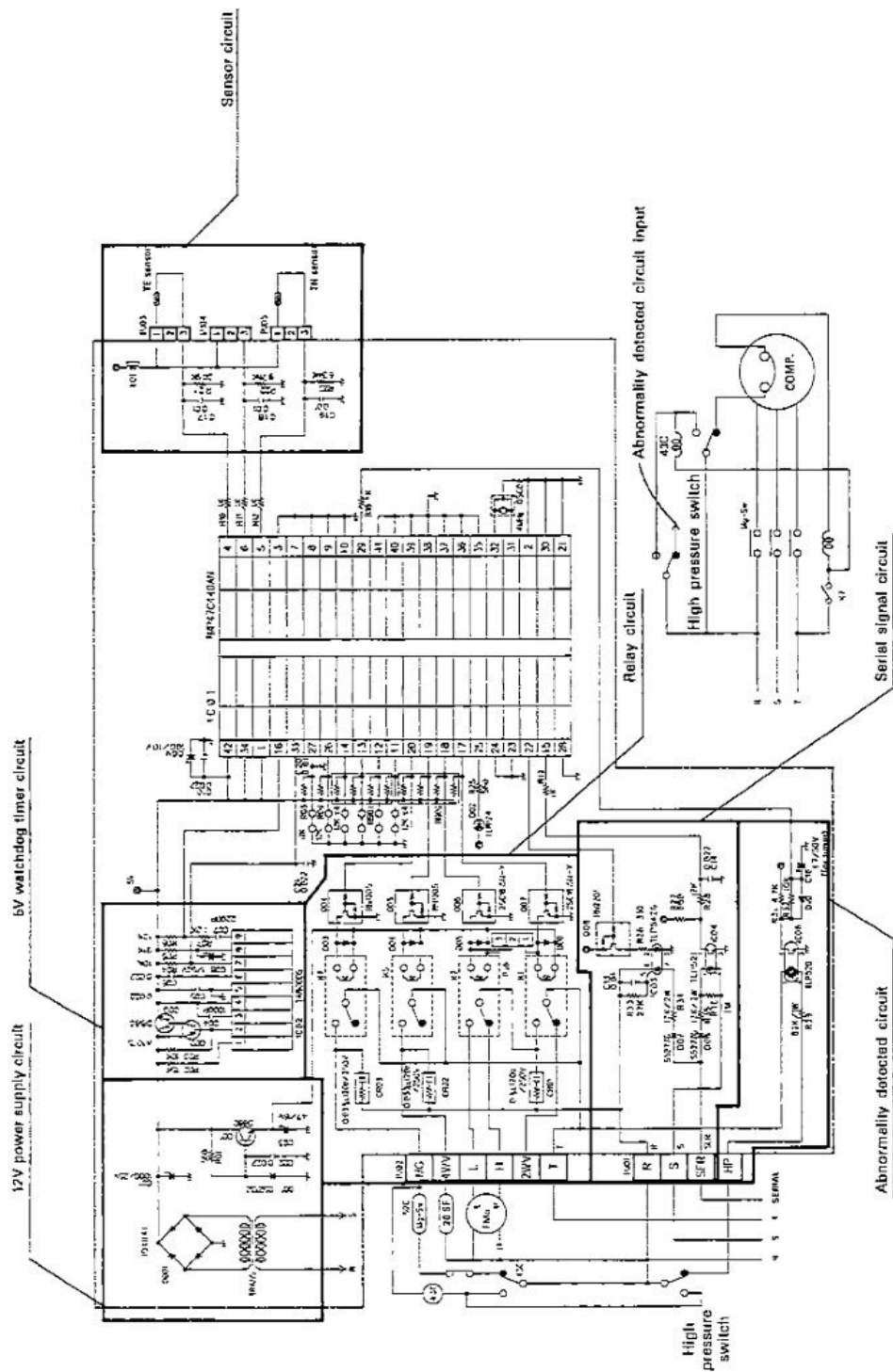


10. DESCRIPTION OF OUTDOOR UNIT CONTROL CIRCUIT

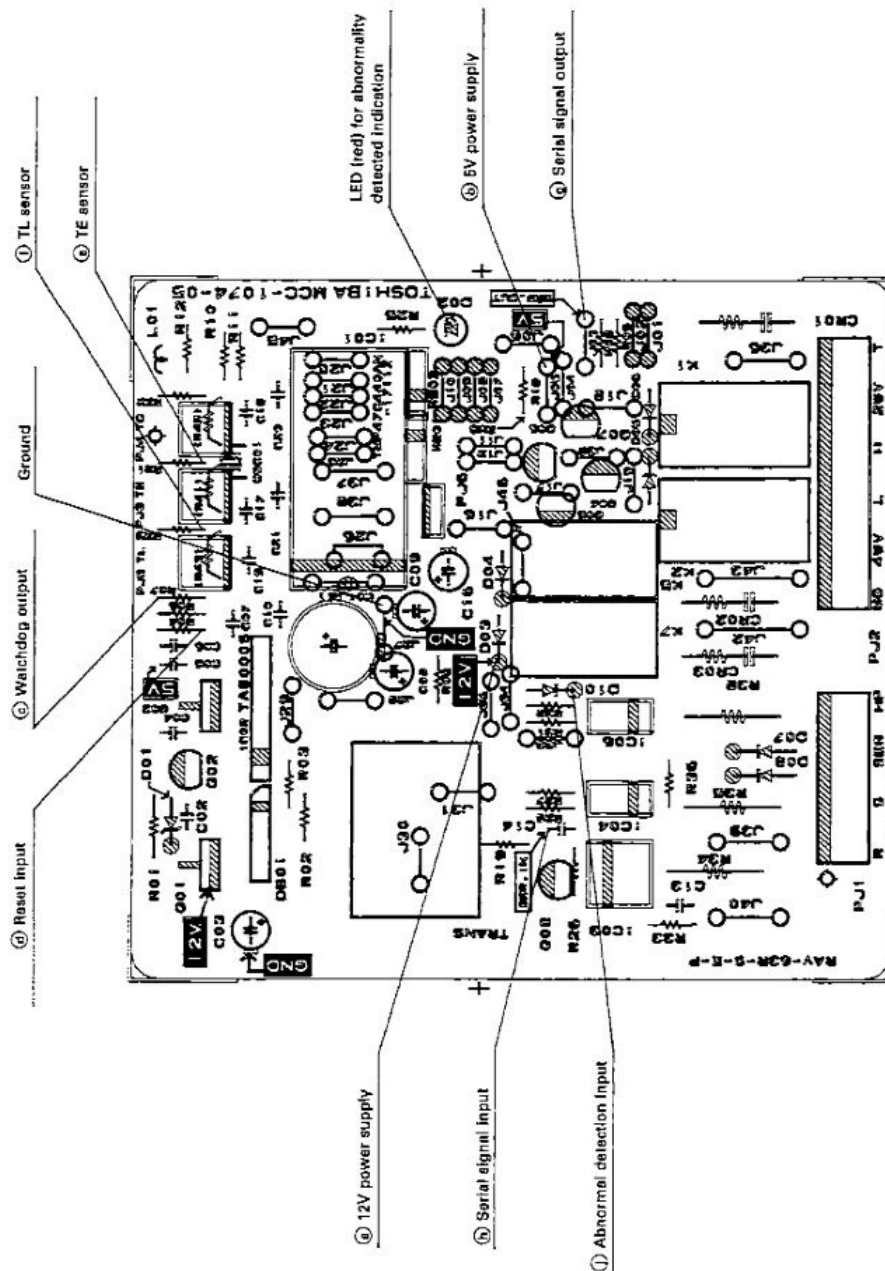
10.1 Outdoor unit control box



10.2 Outdoor unit control circuit diagram

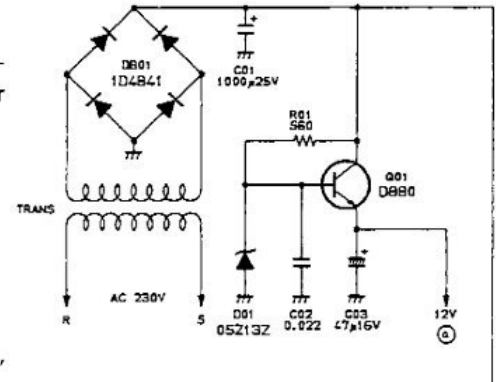


10.3 Outdoor unit PC board



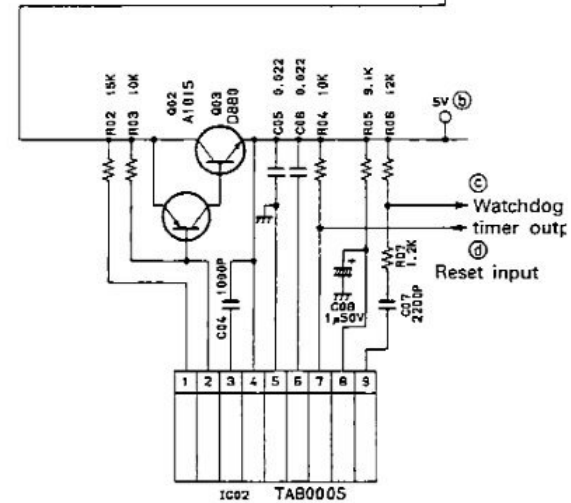
12V power source circuit

Outdoor PC board has a built-in transformer and full-wave rectification by diode bridge (DB01) followed by the provision of transistor (Q01) produces DC power source (a) at 12V.



5V watchdog timer circuit

Basically, the same description as the indoor PC board applies, provided, however, that the reset circuit is not added to the outdoor side.



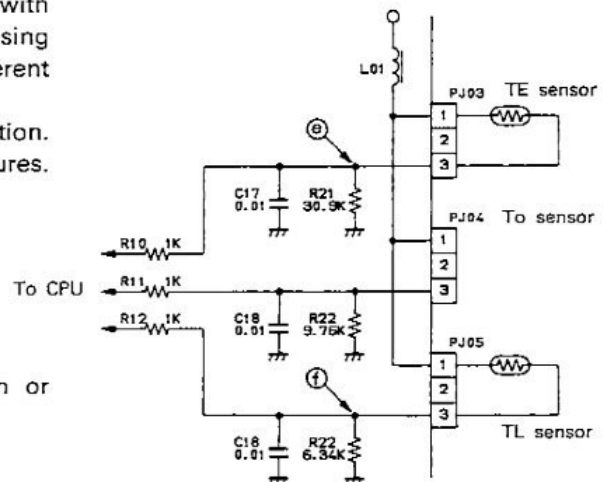
Sensor circuit

This circuit detects the temperature by dividing voltage with resistance and sensor and bringing the voltage value into CPU, using the characteristics of the sensor that resistance varies with different temperatures.

TE is for defrosting, while TL is for low ambient cooling operation. The following voltages are produced at each of the temperatures.

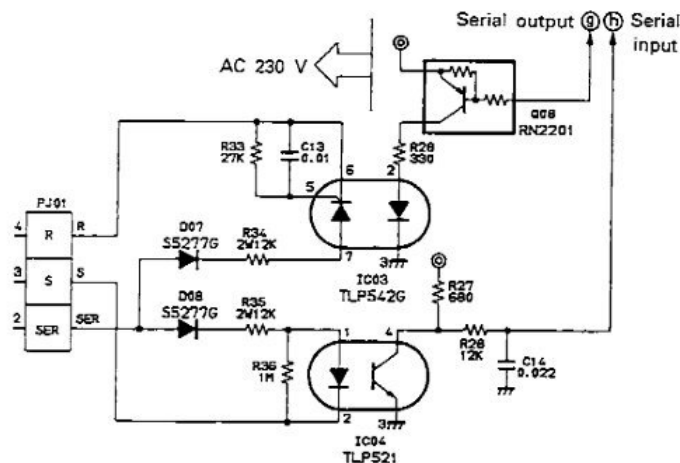
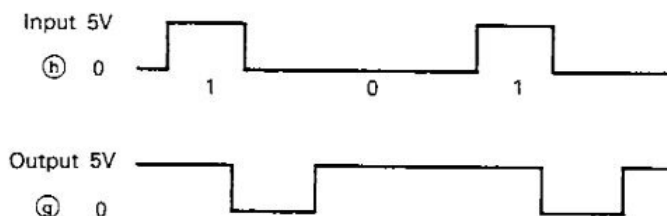
	0°C	25°C
TE (e)	2.3V	3.8V
TL (f)	0.8V	2.0V

When (e) / (f) are at GND or 5V, the sensors are in open or short-circuited.



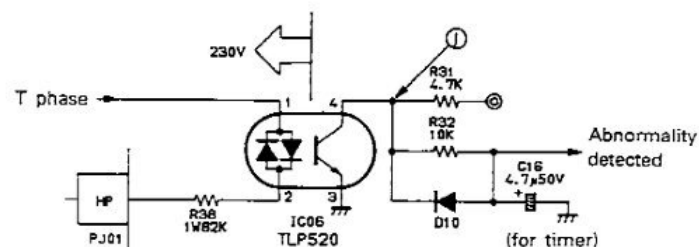
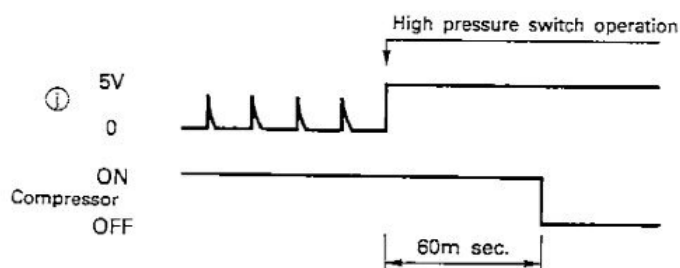
Serial signal circuit (between indoor and outdoor units)

Transmits and receives the signals between indoor and outdoor units in serial signals. As 230V is used for transmitting the signal, the microcomputer section is insulated with photo-coupler with 5V being supplied.



Abnormality-detecting circuit

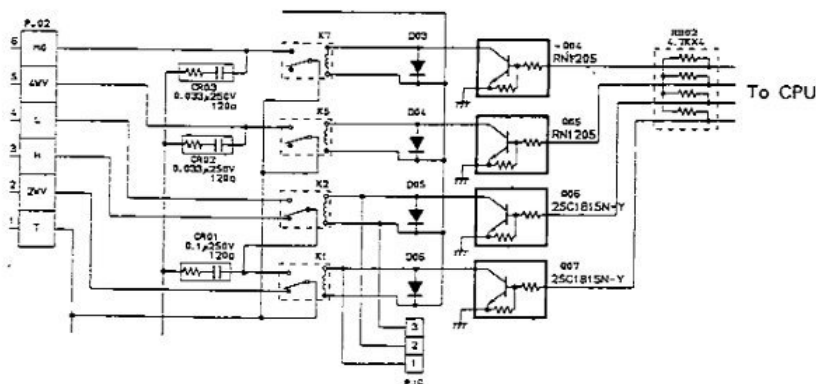
When high pressure switch is operated, abnormality is detected to stop the compressor.



Relay circuit

The relay circuit consists of the diagram in the righthand side.

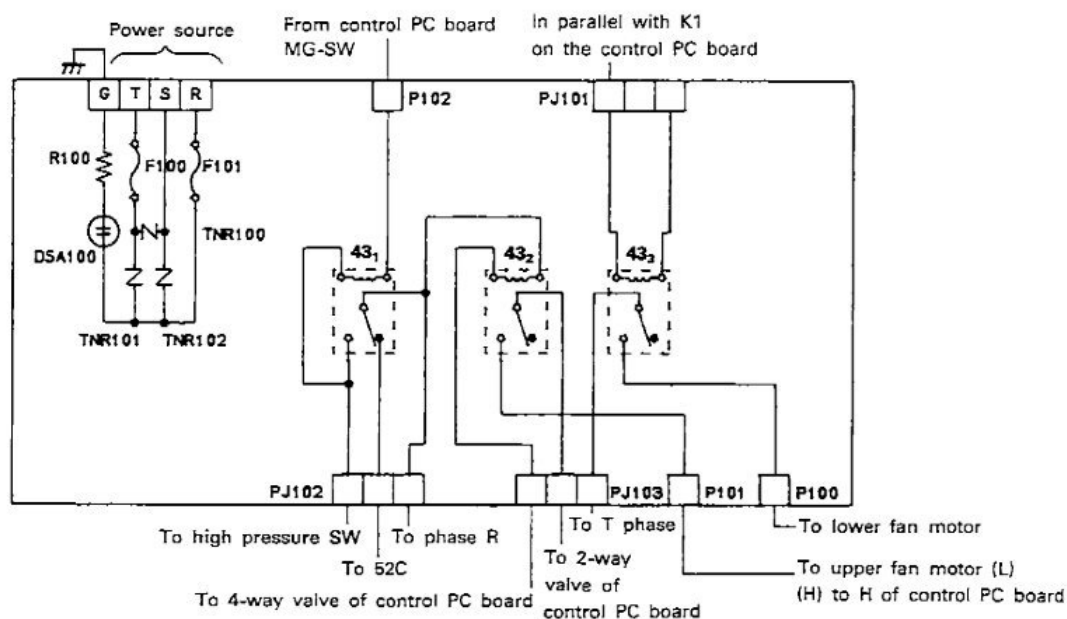
- K1 : Turning fan on and off (2-way valve)
- K2 : Changing over H/L of fan
- K5 : Turning 4-way valve on and off
- K7 : Turning compressor on and off



11. OPERATION OF RELAY PC BOARD (MCC-1075) (RAV-S1003HE8, S1253HE8)

11.1 Purpose: Outdoor fan motor control and absorption of power source surging

11.2 Circuit



11.3 Operation

		K1 OUTPUT	MG SW OUTPUT	4-WAY VALVE OUTPUT	2-WAY VALVE OUTPUT	HIGH PRESSURE SW	FAN MOTOR		
		Relay 43 ₃	Relay 43 ₁	Relay 43 ₂	Relay 43 ₂	Relay 43 ₁	Upper (H)	Upper (L)	Lower
Cooling	Normal operation	○	○	x	x	x	○	x	○
	High pressure switch operation	x	x	x	○ K101 OFF	○	x	x	x
	Low ambient operation	○	○	x	x	x	x K2 ON	x	x
Heating	Normal operation	○	○	○	x	x	○	x	○
	In the release	x	○	○	Through contact 43 ₂	x	x	○	x

12. EMERGENCY OPERATON (COOLING OPERATION ONLY)

By way of temporary expedient, change-over connectors are incorporated which allow for application of 230V directly to indoor fan motor, outdoor fan motor and magnet switch. Drain pump is operated forcibly by pulling out float switch. In this case, operation and stop is effected by ON/OFF of the power line. (The emergency operation is not provided for heating as it can be substituted by other heating appliances and also because of nonavailability of defrosting approach).

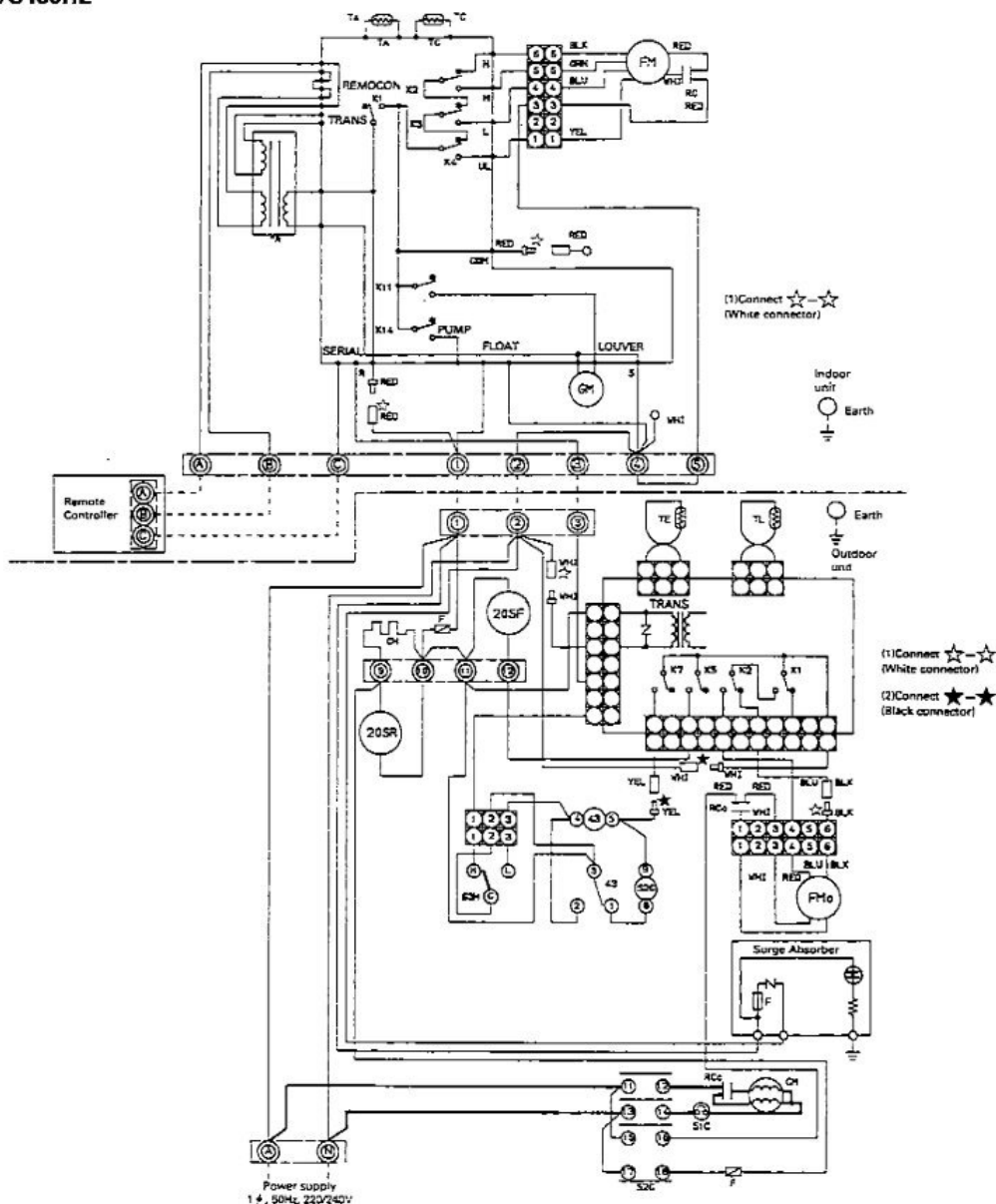
RAV-S1003HE8, S1253HE8

Countermeasures	Indoor connector	Pull out the connector of R phase (red) lead wire from pin (1) and connect it with the connector of lead wire for fan motor K1 output (red).
	Outdoor connector	Pull out #250 Faston (red) lead wire from pin (12) and connect it with the vacant pin (9). Pull out 1P white connector for lower fan motor and connect it with 1P white connector of pin (12). Take out 1P connectors of yellow lead wire and black lead wire for relay output connectors of control PC board and connect yellow lead from relay PC board P102 with black lead coming from (9).
Operation	Operation and stop by the power switch at hand. (High pressure switch becomes the only protective circuit.)	

RAV-S453HE, S713HE8

Countermeasures	Indoor connector	Pull out the connector of R phase (red) lead wire from pin (1) and connect it with the connector of lead wire for fan motor K1 output (red).
	Outdoor connector	Pull out 1P white connector for fan motor and connect it with 1P white connector of pin (2). Take out 1P connectors of yellow lead wire and black lead wire for relay output connectors of control PC board and connect yellow lead from relay 43 [or 43 ₂] with black lead coming (2) [or (6)].
Operation	Operation and stop by the power switch at hand. (High pressure switch becomes the only protective circuit.)	

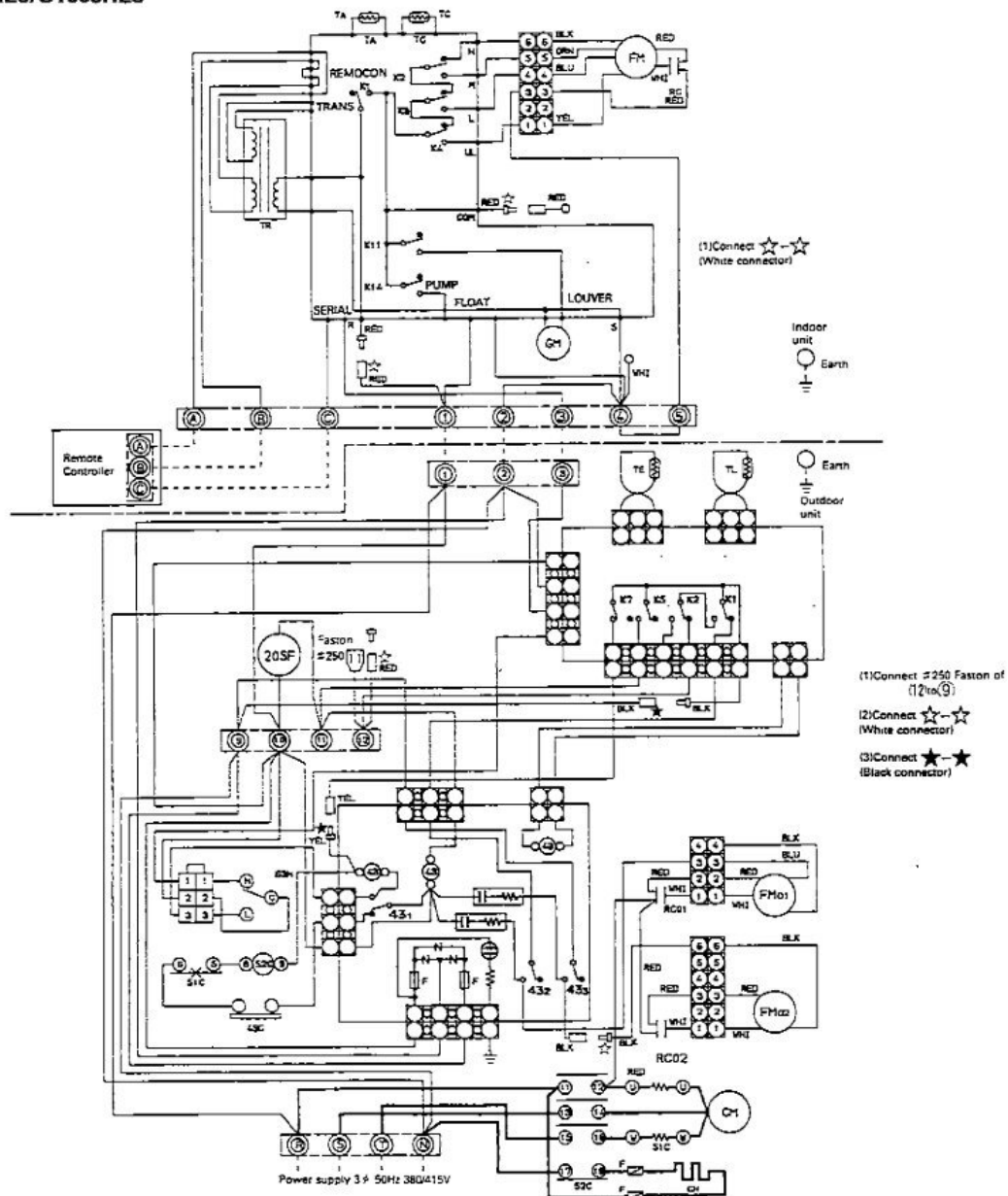
For the method, refer to next page.

RAV-453CHE/S453HE


© shows terminal block and figures show terminal numbers.
Broken lines show wiring at site.

Don't operate the units with the magnetic contactor pushed.

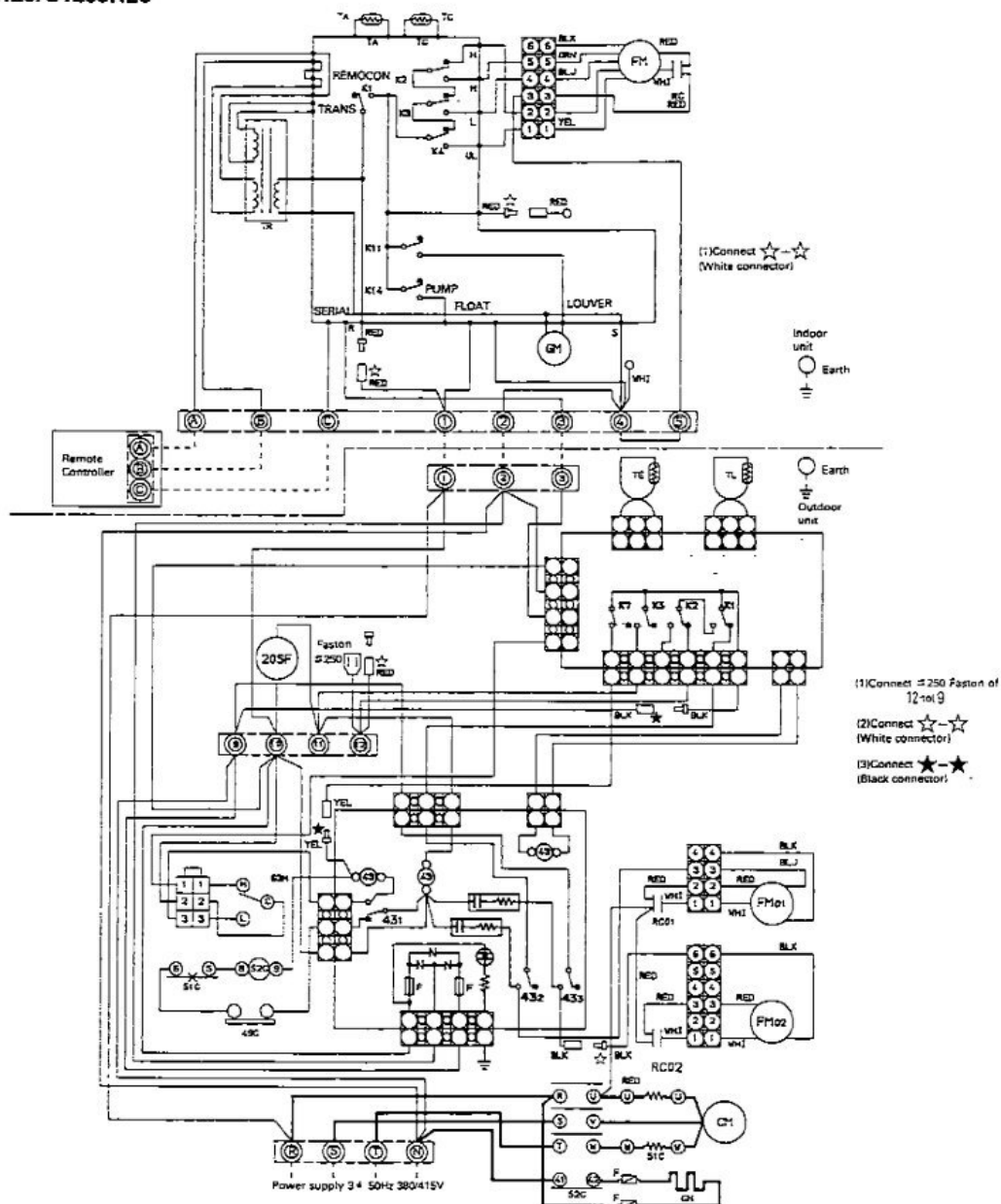
Symbol	Name	Symbol	Name
20SF	Solenoid Coil (4way valve)	CM	Compressor
K ₁ ~K ₁₄	Relay	52C	Magnetic Contactor
20SR	Solenoid Coil (2way valve)	FM _O	Fan Motor (Outdoor)
51C	Overload Relay	TL	Sensor
43	Relay (Outdoor)	TE	Sensor
63H	High Pressure Switch	F	Fuse
CH	Crank Case Heater	RC _O	Running Capacitor (Outdoor Fan Motor)
TA	Sensor	RC _C	Running Capacitor (Compressor)
TC	Sensor	RC	Running Capacitor (Indoor Fan Motor)
FM	Fan Motor (Indoor)		
GM	Induced Draft Motor		
TR	Transformer		

RAV-1003CHE8/S1003HE8


© shows terminal block and figures show terminal numbers.
Broken lines show wiring at site.

Don't operate the units with the magnetic contactor pushed.

Symbol	Name	Symbol	Name
20SF	Solenoid Coil (4way valve)	CM	Compressor
K ₁ ~K ₁₄	Relay	52C	Magnetic Contactor
49C	Inner Overload Relay	FM _{01,02}	Fan Motor (Outdoor)
51C	Overload Relay	TL	Sensor
43 _{1~3}	Relay (Outdoor)	TE	Sensor
63H	High Pressure Switch	F	Fuse
CH	Crank Case Heater	RC _{01,02}	Running Capacitor (Outdoor Fan Motor)
TA	Sensor	RC	Running Capacitor (Indoor Fan Motor)
TC	Sensor	TR	Transformer
GM	Induced Draft Motor		
FM	Fan Motor (Indoor)		

RAV-1253CHE8/S1253HE8


© shows terminal block and figures show terminal numbers.
Broken lines show wiring at site.

Don't operate the units with the magnetic contactor pushed.

Symbol	Name	Symbol	Name
20SF	Solenoid Coil (4way valve)	CM	Compressor
K ₁ ~K ₁₄	Relay	52C	Magnetic Contactor
49C	Inner Overload Relay	FM ₀₁₋₀₂	Fan Motor (Outdoor)
51C	Overload Relay	TL	Sensor
43 ₁ ~ ₃	Relay (Outdoor)	TE	Sensor
63H	High Pressure Switch	F	Fuse
CH	Crank Case Heater	RC ₀₁₋₀₂	Running Capacitor (Outdoor Fan Motor)
TA	Sensor	RC	Running Capacitor (Indoor Fan Motor)
TC	Sensor	TR	Transformer
GM	Induced Draft Motor		
FM	Fan Motor (Indoor)		

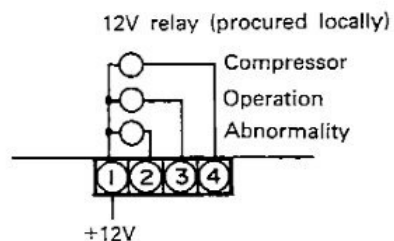
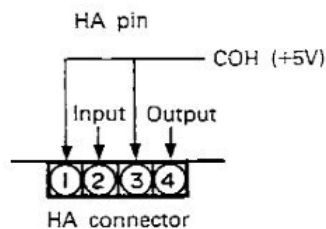
13. APPLIED CIRCUIT

(1) HA terminal
(PJ06)

Connectors that can be used for remote start/stop/display are provided on the indoor PC board.

(2) Display output
(PJ13)

12V output is provided on the indoor PC board for operation display (interlocked with LED), compressor operation display and outdoor abnormality display.

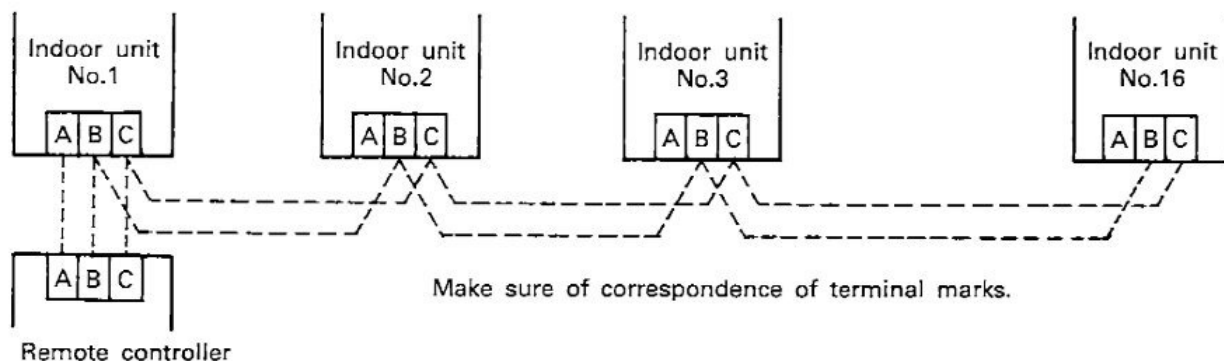


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page 50

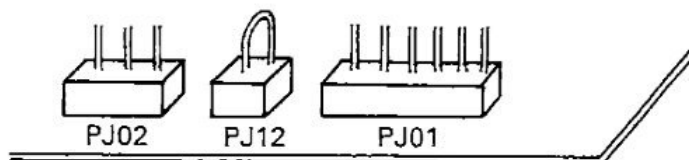
14. WIRING FOR GROUP OPERATION

Up to 16 units can be controlled as a group by one remote controller.

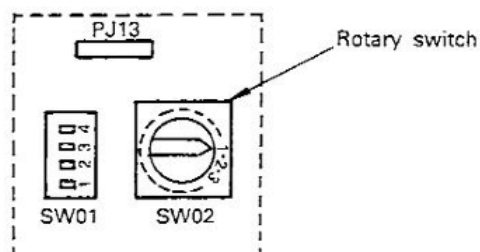
- ① Align the phase sequence of the power supply to all outdoor units.
- ② Connect the terminals A, B, C on both of the remote controller and the indoor unit of No1 unit.
- ③ Connect terminals B, C on both indoor units of No.1 and No.2 unit. Then connect in the same way No.2 and No.3, No.3 and No.4....up to No.16 unit.



- ④ Remove the PJ12-connector on the indoor PC board of No.2 unit and up to No.16 unit to prevent malfunction.

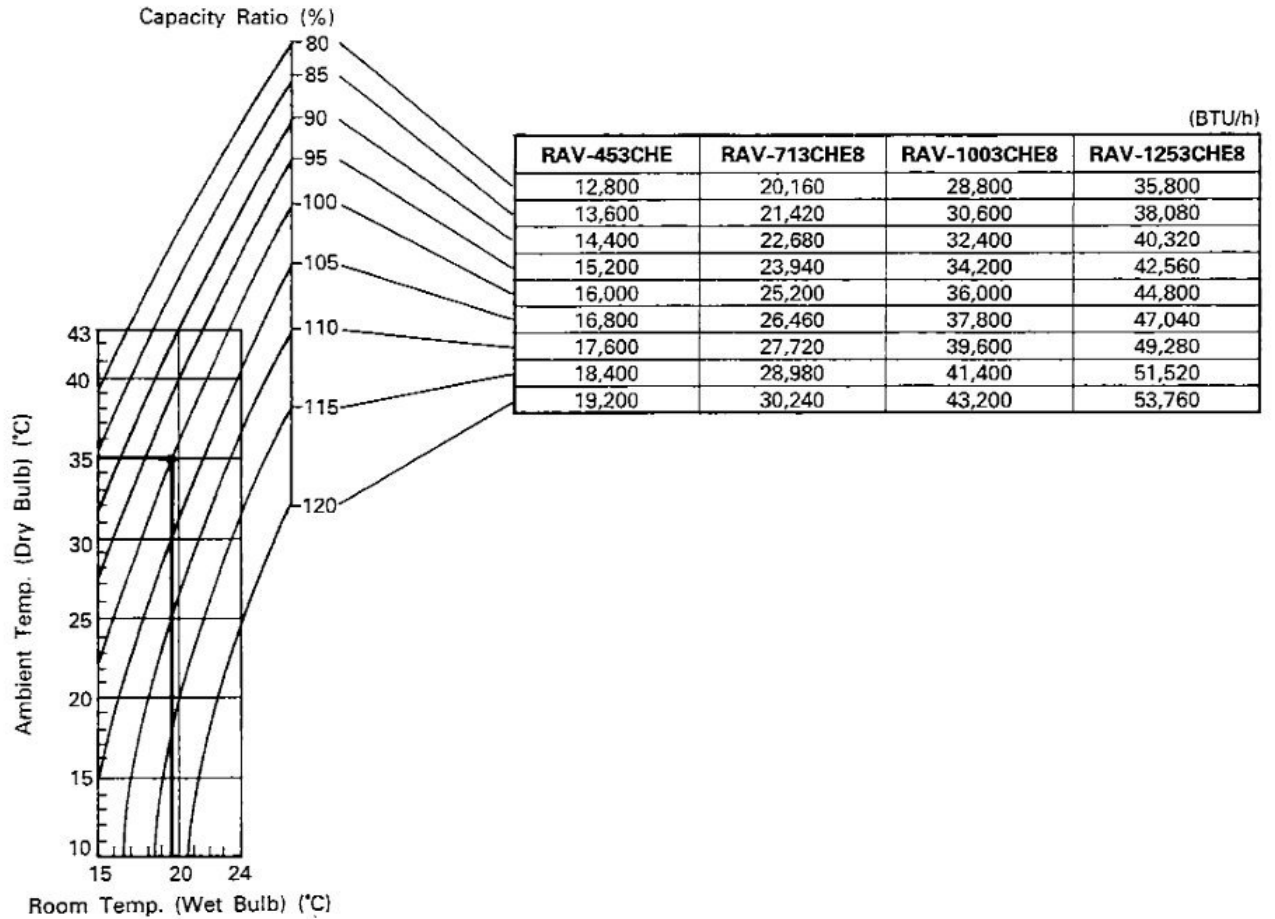


- ⑥ Set each unit No. rotary switch on the indoor PC board. The unit connected to the remote controller should be set as No.1 unit. Then set No.2 and up to No.16 so that start time of each unit is respectively delayed to prevent simultaneous starting current.

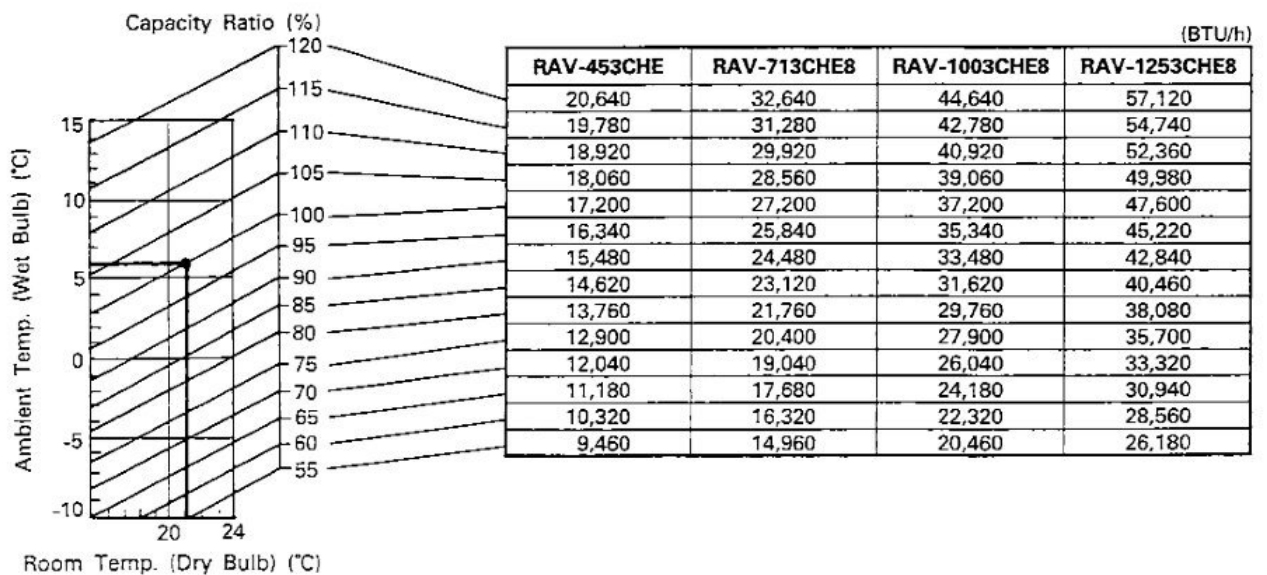


15. PERFORMANCE CHARACTER

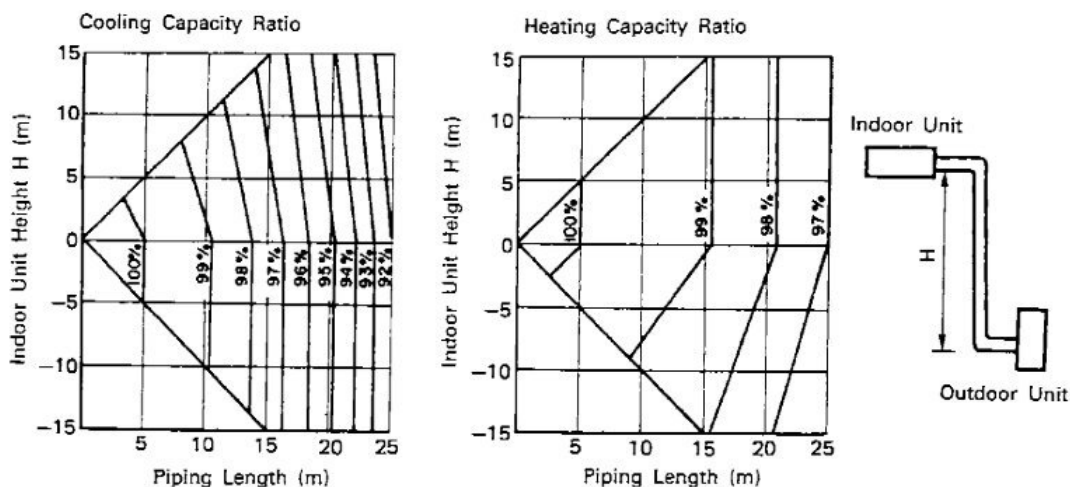
15.1 Cooling capacity



15.2 Heating capacity



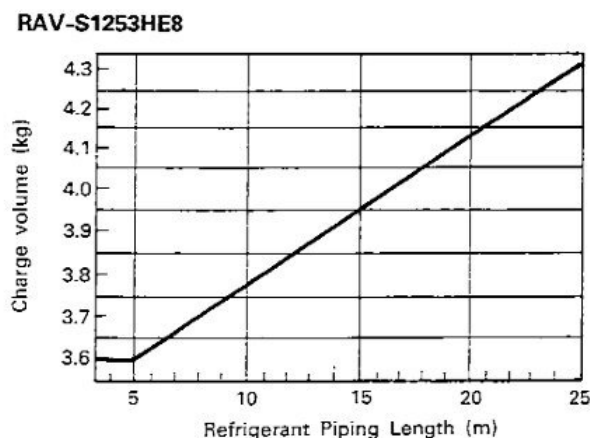
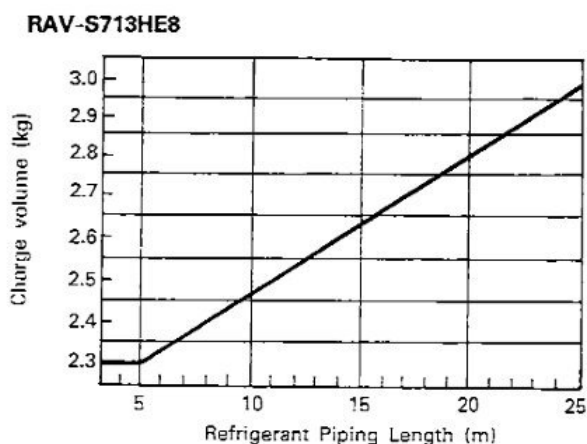
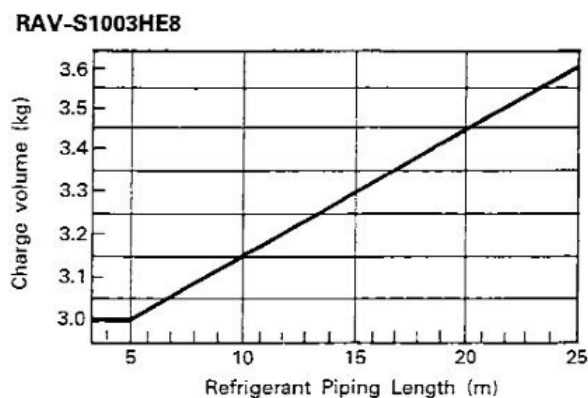
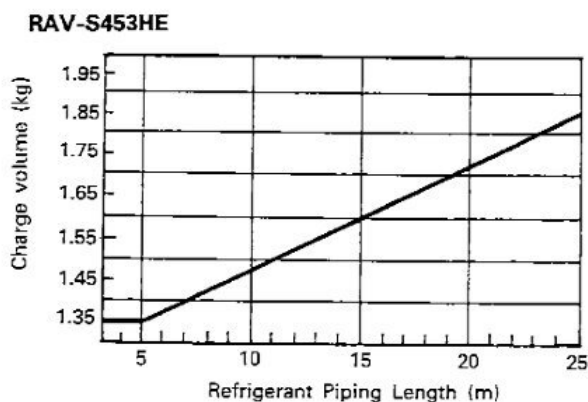
15.3 Piping length/cooling capacity/heating capacity



15.4 Piping length/additional refrigerant volume

In case of piping length is more than 5m, less than 25m.

Model	Add. volume (per 1 meter)	Maximum Add. volume
RAV-S453HE	25 g/m	500 g
RAV-S713HE8	35 g/m	700 g
RAV-S1003HE8	30 g/m	600 g
RAV-S1253HE8	35 g/m	700 g



Notes for flare piping connection

The performance of the split air conditioner depends greatly on the piping work.

Keep following 5 major points for the piping work.

- (1) Do not let dirt and moisture get inside of the pipes
- (2) Perform flaring and connection correctly
- (3) Purge air from the pipes quickly and efficiently
- (4) Check for gas leaks around flare connections
- (5) Add refrigerant gas if the piping length exceeds 5m

Additional supply of refrigerant

On delivery, the outdoor unit contains the refrigerant for one air purging and 5m piping.

The refrigerant must be supplemented, if the piping length exceeds 5m.

However the piping cannot extended beyond 25m.

The amount of refrigerant to be supplemented varies depending on the type of unit.

Refer to the corresponding instruction manual of each unit to determine the amount of refrigerant for each additional meter.

How to calculate the amount of refrigerant to be supplemented (example).

$M = (L-5) \times A \text{ (g)}$	
M :	The amount of refrigerant to be supplemented (g)
L :	The one way actual piping length (m)
5 :	The standard piping length (m)
A :	The additional volume of refrigerant for every 1m (g/m)

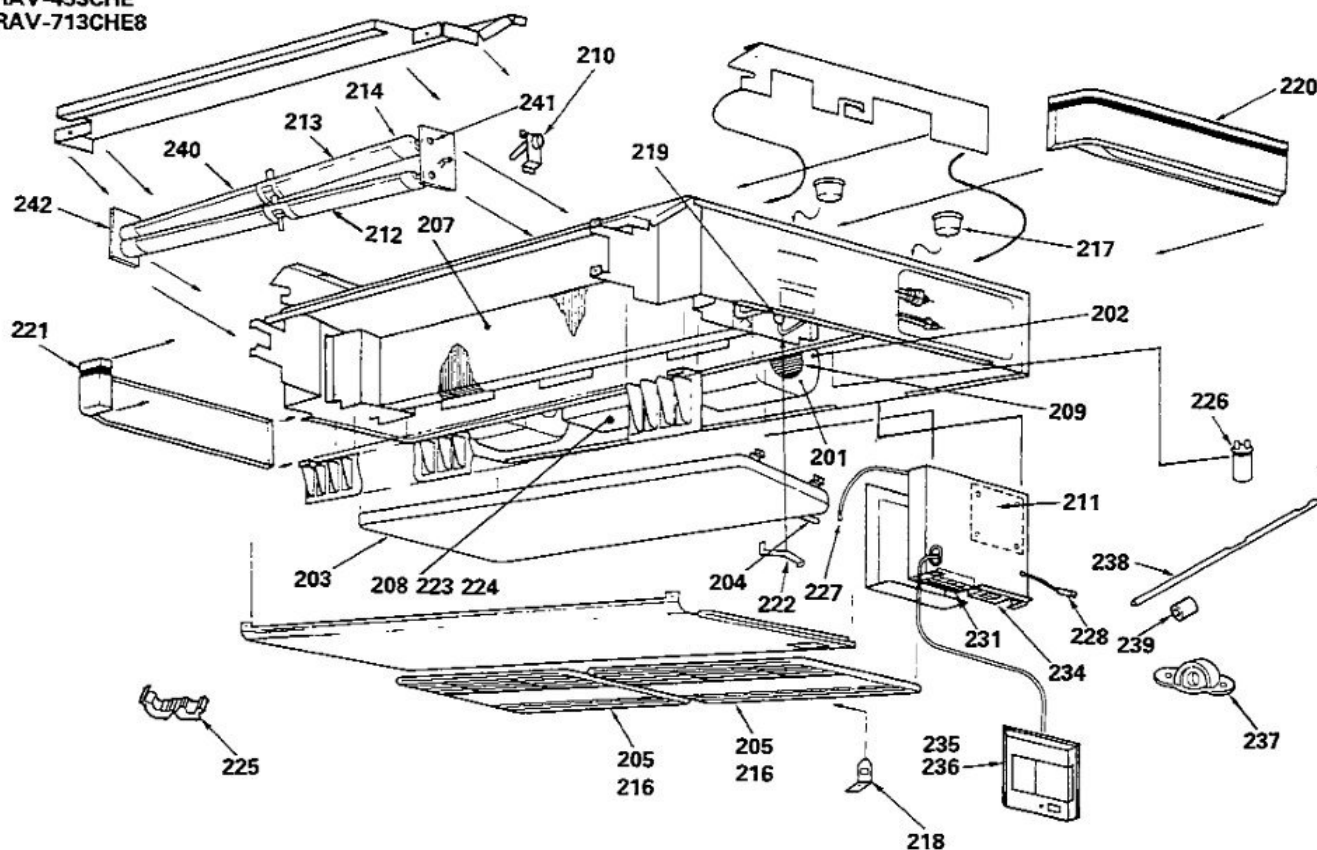
Accuracy of additional supply is $\pm 50\text{g}$.

As an over charge or shortage must give cause for failure of the unit, always measure the precise amount before supply.

16. EXPLODED VIEWS AND PARTS LISTS

16.1 Indoor unit

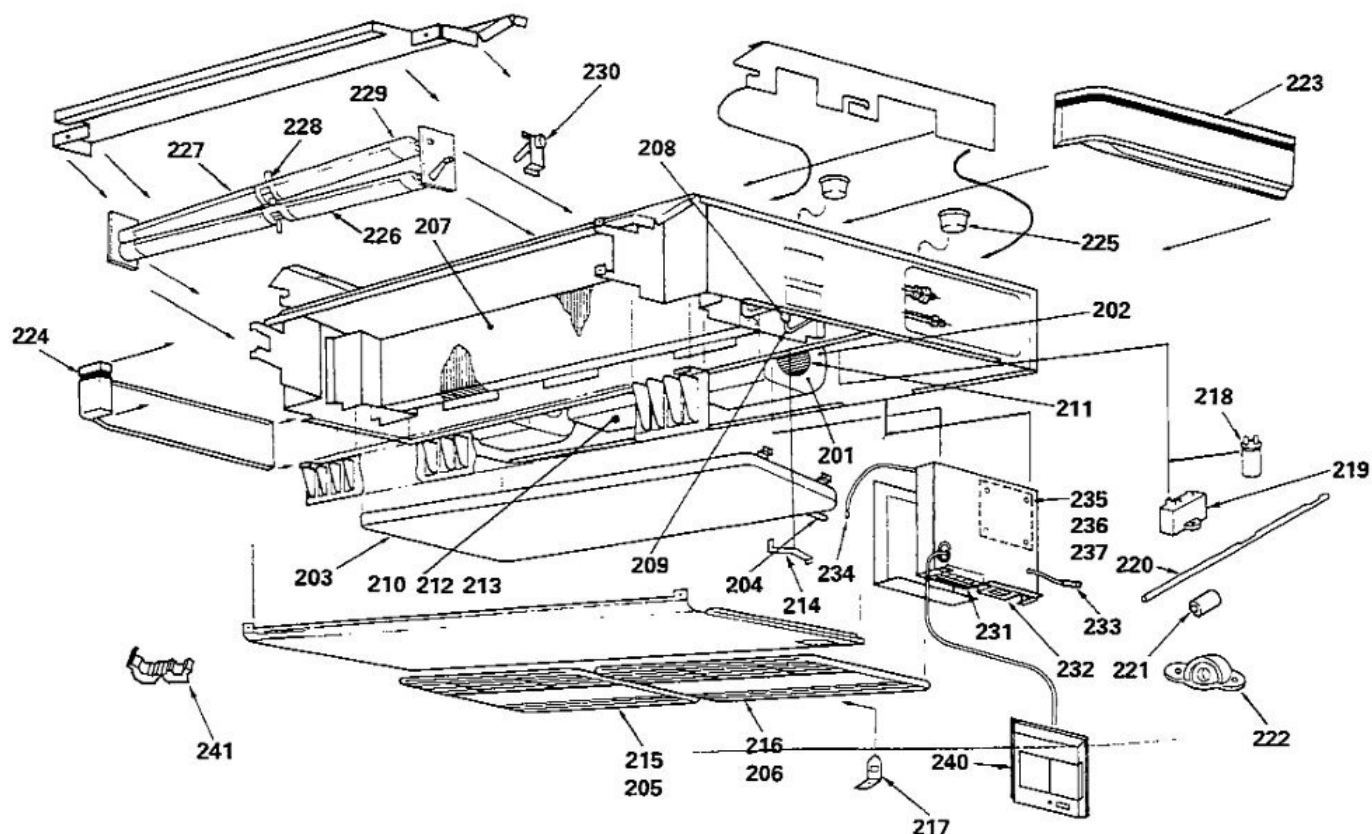
RAV-453CHE
RAV-713CHE8



Location No.	Part No.	Description
201	43126122	Fan Case, Lower
202	43126123	Fan Case, Upper
203	43191214	Drain Pan (713CHE8)
203	43191224	Drain Pan (453CHE)
204	43170183	Hose, Drain
205	43180253	Air-Filter, S
206	43180254	Air-Filter, L
207	43144566	Evaporator (453CHE)
207	43144592	Evaporator (713CHE8)
208	43121452	Motor, AC, 200V, 50Hz, Fan (713CHE8)
208	43121475	Motor, AC, 230V, 50Hz, Fan (453CHE)
209	43120138	Fan, Sirocco
210	43121376	Motor, Air Sweep
211	43169463	PC Board (453CHE)
211	43169470	PC Board (713CHE8)
212	43109249	Grille
213	43109301	Cover, Grille
214	43109306	Support, Grille
215	43109246	Grille, Inlet, S
216	43109245	Grille, Inlet, L
217	43119373	Bolt-Cover
218	43107204	Hinge

Location No.	Part No.	Description
219	43147441	Distributor (453CHE)
219	43047527	Tube, Capillary (713CHE8)
220	43102601	Cabinet, Side, Right
221	43102602	Cabinet, Side, Left
222	43019604	Holder, Sensor
223	43039137	Band, Motor, Right
224	43039136	Band, Motor, Left
225	43063185	Holder, Cord
226	43055292	Capacitor, Electrolytic (453CHE)
226	43155081	Capacitor, Electrolytic (713CHE8)
227	43150118	Sensor
228	43150111	Sensor
231	43160375	Terminal Block, 8P
234	43158094	Transformer, Power
235	43162029	Cover
236	43169462	Remote Controller
237	43125131	Bearing
238	43125132	Shaft
239	43125134	Coupling
240	43109318	Grille, Outlet
241	43109254	Holder, Shaft Right
242	43109255	Holder, Shaft Left

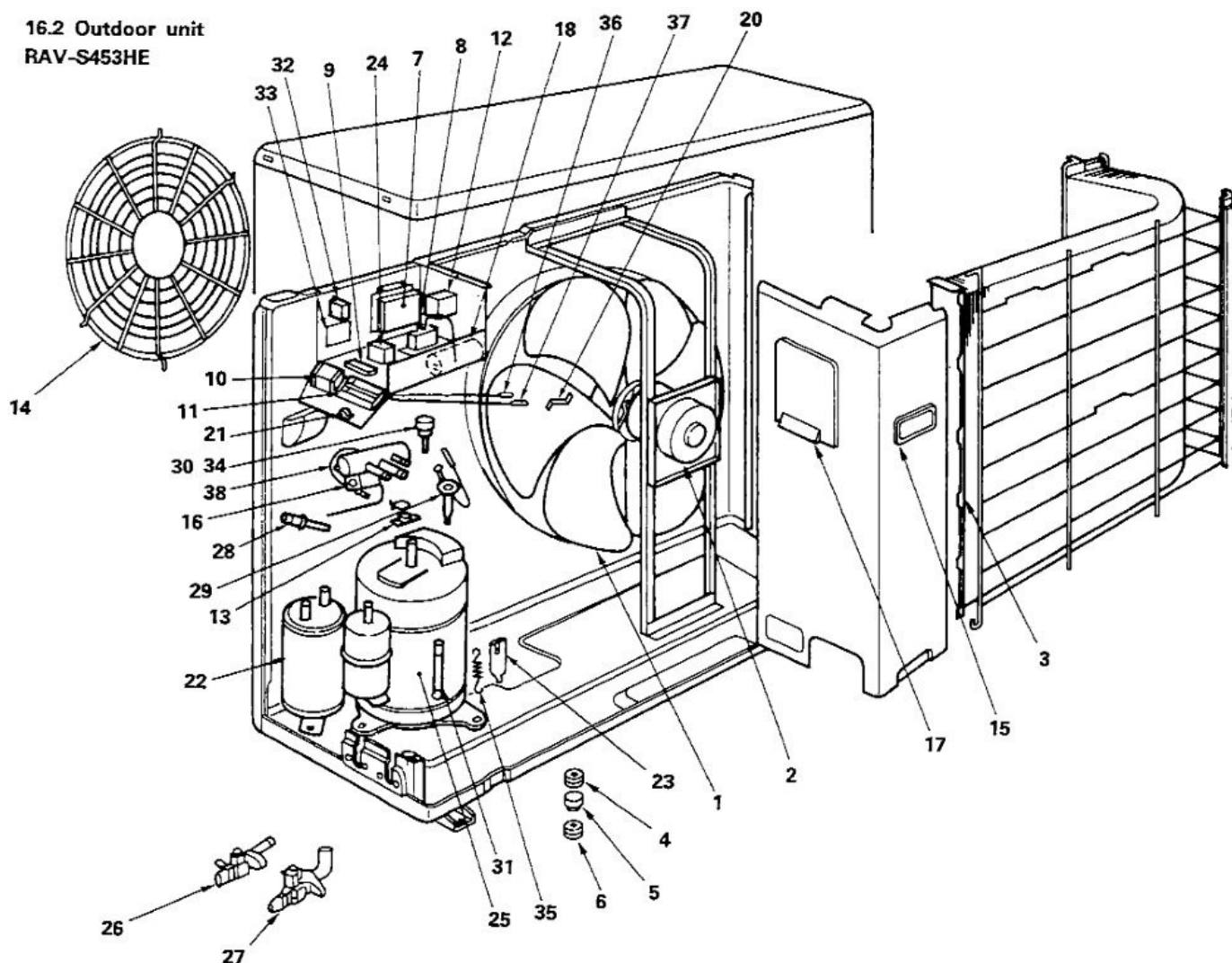
RAV-1003CHE8
RAV-1253CHE8



Location No.	Part No.	Description
201	43126120	Fan Case, Lower
202	43126121	Fan Case, Upper
203	43191215	Drain Pan (1253CHE8)
203	43191217	Drain Pan (1003CHE8)
204	43170183	Hose, Drain
205	43180253	Air-Filter, S
206	43180254	Air-Filter, L
207	43144573	Evaporator (1253CHE8)
207	43144574	Evaporator (1003CHE8)
208	43147509	Distributor
209	43047527	Tube, Capillary
210	43121448	Motor, AC, 230V, 50Hz, Fan (1253CHE8)
210	43121479	Motor, AC, 230V, 50Hz, Fan (1003CHE8)
211	43120095	Fan, Sirocco
212	43039136	Band, Motor, Left
213	43039137	Band, Motor, Right
214	43019604	Holder, Sensor
215	43109246	Grille, Inlet, S
216	43109245	Grille, Inlet, L
217	43107204	Hinge
218	43155064	Capacitor, Electrolytic

Location No.	Part No.	Description
219	43055275	Capacitor, Electrolytic
220	43125136	Shaft
221	43125137	Coupling
222	43125135	Bearing
223	43102601	Cabinet, Side, Right
224	43102602	Cabinet, Side, Left
225	43119373	Bolt-Cover
226	43109244	Grille (1253CHE8)
226	43109248	Grille (1003CHE8)
227	43109304	Accent, Grille (1003CHE8)
227	43109305	Accent, Grille (1253CHE8)
228	43109247	Holder, Grille
229	43109306	Support, Grille
230	43121376	Motor, Air Sweep
231	43160375	Terminal Block, 8P
232	43158094	Transformer, Power
233	43150119	Sensor
234	43150118	Sensor
235	43169463	PC Board
236	43133011	Spark-Killer
237	43133012	Spark-Killer
240	43169462	Remote Controller
241	43063185	Holder, Cord

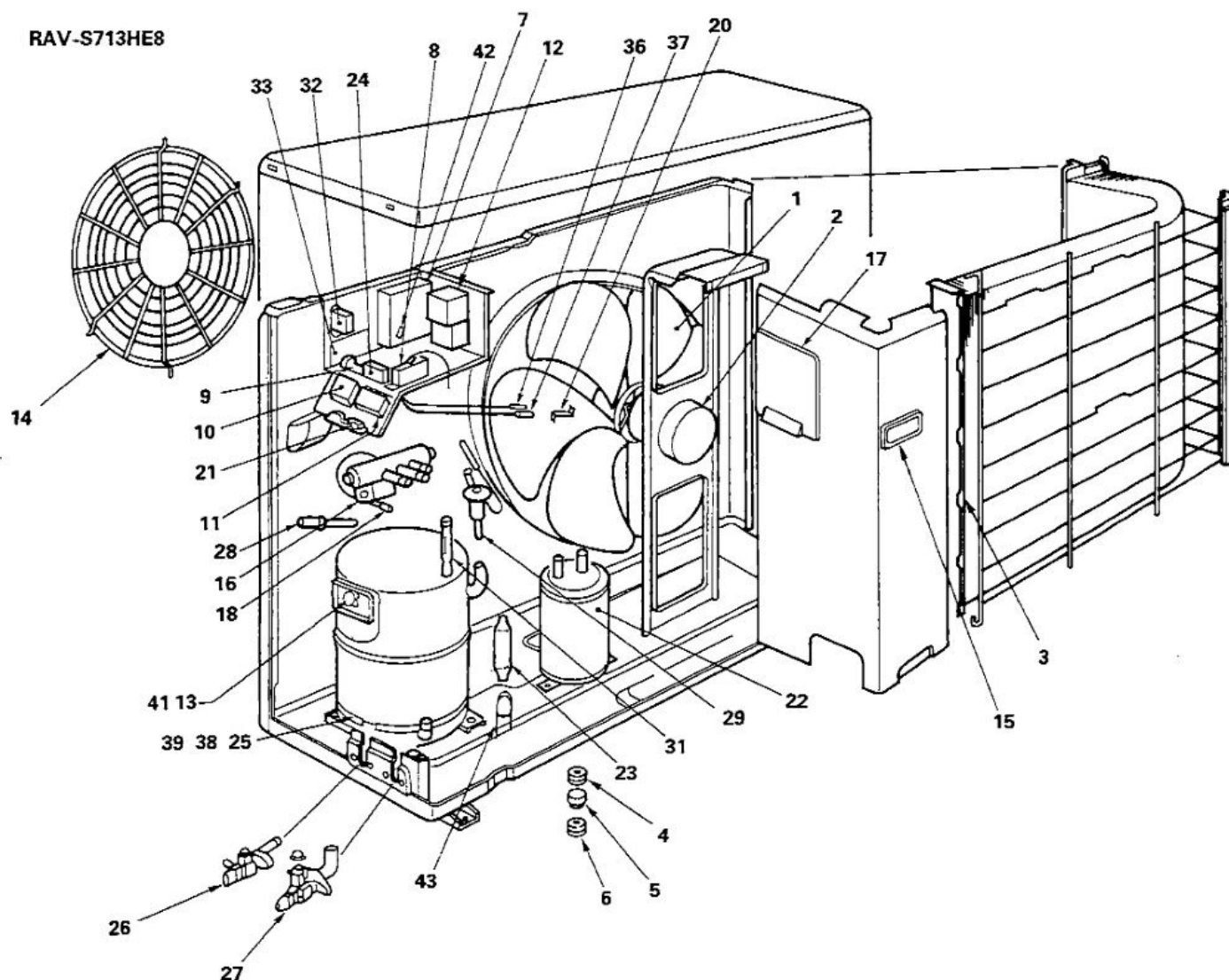
16.2 Outdoor unit RAV-S453HE



Location No.	Part No.	Description
1	43120156	Fan, Propeller
2	43121473	Motor, AC, 230V, 50Hz, Fan
3	43143508	Condenser
4	43149212	Base, Spring, A
5	43149198	Spring, Buffer
6	43149208	Base, Spring, B
7	43169464	PC Board
8	43146387	Switch-High-Pressure
9	43060479	Terminal Block, 4P
10	43160334	Terminal Block, 2P
11	43060750	Terminal Block, 3P
12	43152314	Magnetic, Switch
13	43054286	Relay, Over Load
14	43191252	Guard-Fan
15	43119368	Hanger
16	43045137	Solenoid Coil
17	43162027	Cover, Electric Parts
18	43055366	Capacitor, Plastic Film, 45MFD, 420V
20	43019604	Holder, Sensor

Location No.	Part No.	Description
21	43063114	Holder
22	43148105	Accumulator
23	43145082	Dryer
24	43155047	Capacitor, Electrolytic
25	43041837	Compressor, AC, 220/240V, 50Hz, PH230X3-4LS
26	43146405	Packed Valve, 1/4 in.
27	43146406	Packed Valve, 1/2 in.
28	43147321	Check Joint
29	43146424	Expansion Valve
30	43046198	Coil, 2 Way Valve
31	43146283	Check Valve
32	43153003	Relay (SA106H-230)
33	43169465	Surge Absorber
34	43046151	2 Way Valve
35	43193043	Spring
36	43150130	Sensor, Cond, Out
37	43150129	Sensor, Heat, Exch.
38	43046009	4 Way Valve

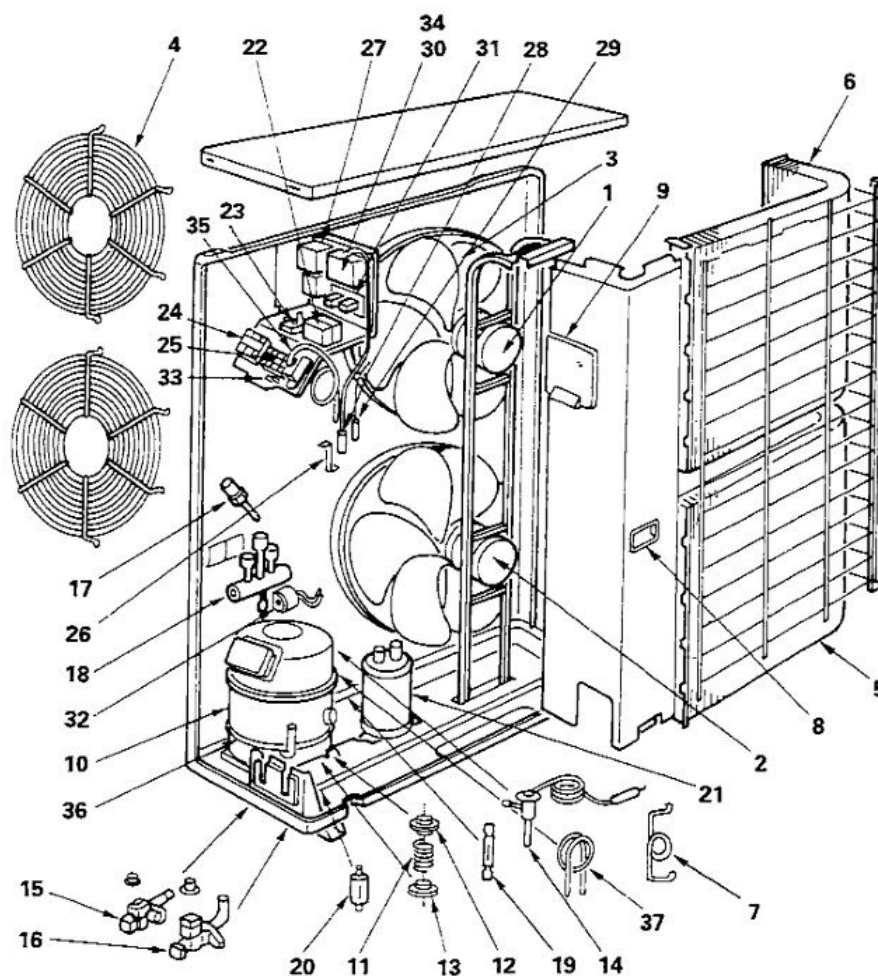
RAV-S713HE8



Location No.	Part No.	Description
1	43120129	Fan, Propeller
2	43121489	Motor, Fan
3	43143505	Condenser
4	43049132	Base, Spring, B
5	43149198	Spring, Buffer
6	43149212	Base, Spring, A
7	43169469	PC Board
8	43146387	Switch-High-Pressure
9	43060157	Terminal Block 4P
10	43160264	Terminal Block
11	43060750	Terminal Block 3P
13	43050244	Thermostat, Bimetal
14	43191252	Guard-Fan
15	43119368	Hanger
16	43046137	Solenoid Coil
17	43162027	Cover, Electric Parts
18	43046009	4 Way Valve
20	43019604	Holder, Sensor
21	43063114	Holder

Location No.	Part No.	Description
22	43148096	Accumulator
23	43145092	Dryer
24	43155080	Capacitor, Plastic Film
25	43140213	Compressor, YH277JA
26	43146415	Packed-Valve, 3/8 in.
27	43146417	Packed-Valve, 5/8 in.
28	43147321	Check Joint
29	43146291	Expansion Valve
31	43146283	Check Valve
32	43153003	Relay (SA106H-230)
33	43169438	Surge Absorber
36	43150130	Sensor, Cond. Out
37	43150129	Sensor, Heat Exch.
38	43193043	Spring
39	43157097	Crankcase, Heater
41	43169254	Holder, Thermostat
42	43060229	Fuse, 1A
43	43146432	Tube Capillary

RAV-S1003HE8
RAV-S1253HE8



Location No.	Part No.	Description
1	43121463	Motor, AC, 230V, 50Hz, Fan
2	43121464	Motor, AC, 230V, 50Hz, Fan
3	43120129	Fan, Propeller
4	43191252	Guard-Fan
5	43143389	Condenser, Lower (S1253HE8)
5	43143478	Condenser, Lower (S1003HE8)
6	43143513	Condenser, Upper (S1253HE8)
6	43143514	Condenser, Upper (S1003HE8)
7	43047492	Capillary-Tube (S1003HE8)
7	43047527	Tube, Capillary (S1253HE8)
8	43119368	Hanger
9	43162027	Cover, Electric Parts
10	43140404	Compressor, YH406JA (S1003HE8)
10	43140506	Compressor, YH506JA (S1253HE8)
11	43149198	Spring, Buffer
12	43149212	Base, Spring, A
13	43149208	Base, Spring, B
14	43146362	Expansion Valve
15	43146415	Packed Valve
16	43146416	Packed Valve
17	43147386	Check Joint
18	43046008	4 Way Valve (S1253HE8)

Location No.	Part No.	Description
18	43146251	4 Way Valve (S1003HE8)
19	43146283	Check Valve
20	43145092	Dryer
21	43148096	Accumulator
22	43155047	Capacitor, Electrolytic
23	43060479	Terminal Block, 4P
24	43160264	Terminal Block
25	43060324	Terminal Block, 3P
26	43019604	Holder, Sensor
27	43152312	Magnetic-Switch (S1253HE8)
27	43152329	Magnetic Contactor, MF-20FR1-FS (S1003HE8)
28	43150129	Sensor, Heat Exch.
29	43150130	Sensor, Cond. Out
30	43169469	PC Board
31	43169467	PC Board
32	43046072	Solenoid Coil
33	43063070	Holder, Cord
34	43163016	Support
35	43146387	Switch-High-Pressure
36	43157140	Heater, Crankcase
37	43146430	Capillary-Tube (S1003HE8)
37	43146431	Capillary-Tube (S1253HE8)

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